

High precision segmentation of glioma and surroundings – a feasibility study using multiparametric MRI and deep learning

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Abstract

Glioblastoma (GBM) is an aggressive primary brain cancer in which precise spatial characterization of tumor sub-compartments and surrounding anatomy may support research into targeted treatment. We designed and assessed a feasibility pipeline for subject-specific localization of glioma and brain regions from standard multiparametric MRI (T1, T1-Gadolinium, T2, FLAIR), combining deep-learning segmentation with robust anatomical parcellation rather than atlas coregistration. Coregistered images yield three tumor compartments: central non-enhancing/necrotic tumor, enhancing tumor, and surrounding edema. From this multichannel representation, we derive tumor volumes and regional tumor burden “hit-plots” that show, at the voxel level, which brain regions each compartment intersects. In a cohort of $n = 50$ UCSF-PDGM glioblastoma subjects, deep-learning segmentations agreed closely with the reference masks: median Dice of 0.90 (whole tumor), 0.94 (tumor core), and 0.86 (enhancing tumor), with corresponding HD₉₅ values of 4.1, 2.2, and 2.0 mm. In one longitudinal LUMIERE subject (Patient-048, six timepoints), the pipeline recovered concordant volumetric trajectories and evolving regional-burden patterns with respect to specific brain structures, in agreement with independent comparator segmentations. This subject-specific regional tumor profile may in the future serve as a descriptive anatomical context for radiotherapy target delineation.

Keywords: Glioblastoma; Neuro-oncology; MRI; Tumor segmentation; Deep learning

1 Introduction

Glioblastoma (GBM) is a highly aggressive brain cancer accounting for ~50% of all primary malignant brain tumors; it is more frequent in men than women, and the mean age at diagnosis is ~65 years [39, 85]. Glioblastoma is defined molecularly as *IDH-wildtype* [65, 87, 102].

Even with maximal multimodal therapy, the median overall survival is only ~9–12 months from diagnosis, and five-year survival is below 5% [39, 85, 14]. Macroscopically detectable GBM in adults occurs most frequently in the cerebral hemispheres and most often in the temporal and frontal lobes of the brain, with typical migration of malignant cells into adjacent brain tissue (diffuse glioma) [56].

Glioma-neuron interactions through synaptic remodelling [55, 46], and the role of cognitive functioning as a predictor of survival in newly diagnosed high-grade glioma, especially in older patients [52, 98, 54], further motivate per-region anatomical profiling that respects the patient’s own neuroanatomy.

Another line of research on the spatial location and extent of tumor growth concerns recent advances in radiotherapy, precision imaging, target delineation, and updated field guidelines. Radiotherapy (RT) is well-established as a core treatment modality in the management of glioblastomas [99, 100]. The development of more accurate and adaptive radiotherapy planning and delivery methods such as intensity-modulated radiotherapy (IMRT) and volumetric modulated arc therapy makes it possible to sculpt dose distributions around critical brain structures, minimizing the volume of normal brain receiving moderate to high radiation doses [76, 77].

Alongside these developments, precision imaging techniques such as high-resolution multiparametric magnetic resonance imaging (mpMRI) [31, 32, 76, 91, 94] have become increasingly crucial for target delineation, with deep-learning methodologies broadly adopted [103] and contributed to in our own work [4, 40, 42, 66, 50].

The heterogeneity of GBM is also reflected in its imaging features or phenotypes [2, 49, 83]. Different biological properties of glioblastomas can be reflected in various tumor subregions [34], such as enhancing tumor (ET) and non-enhancing tumor (NET) regions; peritumoral brain regions that represent edematous and invaded tissue (ED), as well as central or tumor core (TC) regions, encompassing non-enhancing and enhancing regions. These subregions can exhibit different signal intensity characteristics in different MRI measurement techniques, i.e., spatiotemporal heterogeneity in multiparametric MRI [78]. Recent MRI-radiomics work has further explored this avenue: for example, [51] build an MRI radiomics signature for non-invasive prediction of 1p/19q codeletion status in lower-grade glioma, and [58] use a transfer-learning approach on MRI radiomics to predict overall survival across lower- and high-grade gliomas. The present work is complementary: instead of summarising the tumor as a single radiomics vector, we keep the per-region structure explicit through the Hit-Plot variants introduced in the following.

Of particular relevance to our work is the recently updated ESTRO-EANO consensus on clinical target volume (CTV) delineation in adult glioblastoma [76, 41], in which the gross tumor volume (GTV) is expanded by a 15 mm CTV margin (reduced from 20 mm pre-2023) and a further ~ 3 mm planning target volume (PTV) margin, while explicitly accounting for organs at risk (OAR) [72] and normal-tissue complication probability (NTCP) models [77]. A detailed treatment of GTV/CTV/PTV definitions, of the FLAIR-versus-edema distinction in CTV adaptation, and of NTCP modelling is given in the cited guideline papers; here we treat these guidelines as the clinical context that motivates per-region anatomical profiling rather than as material to re-derive in the Introduction.

In this new neuro-oncological “landscape” of oncological neuroscience, precision imaging, and new radiotherapy guidelines, [28] argue that an anatomical fingerprint is important for non-invasive assessment of patient prognosis and for surgical and radiotherapy planning context. Several studies have reported anatomical distribution and localisation of glioblastomas at progressively finer spatial granularity. Altieri and coworkers [6] used per-lobe involvement scored by neurosurgeons on T1c MRI and found that frontal localisation was associated with IDH1 mutation (better prognosis), temporal involvement with IDH1 wild-type, parietal location with MGMT methylation, and insular involvement with unmethylated MGMT. Li and coworkers [61] extended this to 990 supratentorial diffuse gliomas, mapped onto the MNI-152 template [69] with the Desikan-Killiany cortical atlas [29], and reported interactions among age, pathological type, and anatomic location at finer granularity. Most recently, Bao and coworkers [9] constructed a population-based GBM/LGG frequency map after MNI registration, used spatial correlation with PET-derived neurotransmitter maps via `JuSpace` [30] as additional support for the neuron-glioma interaction hypothesis of [55], and trained a location-based survival classifier that reached 71% 1-year accuracy on their test set.

Recent clinical and research systems have substantially advanced automated glioma segmentation and volumetry. FDA-cleared tools such as NS-HGlio [1, 95] and commercial brain-tumor analysis platforms [27, 97, 96] provide semi-automatic segmentation, visualization, and longitudinal volumetric quantification for high-grade glioma in clinical workflows. Research tools such as `mri_TumorSynth` [105, 80, 36] and foundation-model approaches such as `BrainIAC` [89, 3] point to increasingly generalizable segmentation backbones. These developments strengthen the case for automated tumor delineation, but most are aimed primarily at voxel-level segmentation or longitudinal volume measurement. The present work instead asks how tumor compartments intersect the patient’s own detailed neuroanatomy in native space.

Closest in spirit to the present work is the `GSI-RADS/Raidionics` line of open-source pipelines [15, 17, 16, 43, 53], which combine deep-learning tumor segmentation with diffeomorphic registration via `SyN/ANTs` [7] to a standardised MNI atlas [37] and emit per-subject reports of laterality, volume, multifocality, resectability, and cortical / subcortical location profiles. The crucial methodological difference (and the direct motivation for the Hit-Plot variants introduced below) is that `Raidionics` expresses location features in atlas (MNI) space, whereas the present pipeline computes them in the patient’s own *native* space and explicitly preserves per-region structure rather than collapsing to scalar atlas-coordinate descriptors.

Our contribution represents a feasibility study of precision imaging in brain cancer. It is developed and evaluated on a leak-safe cohort of $n = 50$ IDH-wildtype glioblastoma subjects from the public UCSF-PDGM dataset, together with a longitudinal LUMIERE case (Patient-048, six timepoints), with varying anatomical locations and extents of glioma. Our work is motivated by new insights into oncological neuroscience, updated radiotherapy guidelines, and advances in medical imaging. Combining sophisticated image segmentation tools with multiparametric MRI allows us to analyze multicompartment tumor features and surrounding tissue with enhanced spatial precision. We report segmentation and native-space anatomical profiling only. Prognostic assessment, radiotherapy planning, and patient outcomes are not evaluated here.

Contributions. This work contributes to *anatomy-aware glioma profiling in native subject space* along three axes. **(i) The Hit-Plot and its variants.** We introduce the Hit-Plot, a per-subject overlap matrix between three tumor compartments (NCR, ED, ET) and the 172 anatomical regions from `SynthSeg/wmparc`, computed in the patient’s own subject space without atlas warping. The implemented variants are a volumetric Hit-Plot, a probabilistic Hit-Plot that propagates the segmenter’s per-class sigmoid probabilities into per-cell mean and standard deviation, and a compact cohort-level functional-anatomy summary that groups the same `wmparc` cells into lobar, motor-adjacent, left language-adjacent, and deep-midline / commissural label sets. Cortical-surface, tract / organ-at-risk proximity, and Hit-Plot-to-outcome variants remain named future work in the *Limitations and translational scope* subsection of the Discussion. **(ii) A unified single-engine segmentation backbone.** We adopt a single state-of-the-art pretrained backbone, the MONAI Bundle BraTS three-compartment SegResNet, and run it on all 50 UCSF-PDGM subjects and on the LUMIERE longitudinal subject (Patient-048, six timepoints; Figure 6) behind a thin Python adapter that decouples the choice of backbone from downstream Hit-Plot code. Dataset-provided masks are retained as a per-subject *reference*, not as *ground-truth*, and the agreement of Hit-Plot computed from the provided masks (Hit-Plot-GT) versus from the unified deep-learning masks (Hit-Plot-DL) is reported per cell as a quantitative measure of the Hit-Plot’s sensitivity to segmentation variability. **(iii) A reproducible n=50 evaluation infrastructure.** The leak-safe cohort (Table 1) and the deterministic build chain (single random seed, pinned subject list, fixed software stack, build chain reproducing every figure and table) together make the framing in axes (i) and (ii) reproducible and

falsifiable on the same data. See *Code and data availability*.

2 Materials and Methods

We first describe the UCSF-PDGM dataset and the selection of cases we have used, followed by the longitudinal examination of a case (Patient-048) from the LUMIERE dataset [88]. Next, we provide details about our processing pipeline.

The study was conducted in accordance with relevant ethical guidelines and regulations. For the UCSF-PDGM dataset, data collection followed relevant guidelines and regulations, and the study was approved by the University of California San Francisco institutional review board with a waiver for consent as described in the original publication. The LUMIERE dataset was collected following ethical approval from the appropriate institutional review board at the University Hospital of Bern, Switzerland, with details provided in the published dataset description. All patient data were anonymized prior to analysis in accordance with standard research protocols. No additional ethical approval was required to analyze these publicly available, anonymized datasets.

Terminology To avoid the interchangeable use of *segmentation accuracy*, *anatomical localization*, and *anatomical profiling*, we fix the following operational definitions and use them consistently throughout the manuscript.

(i) Segmentation accuracy is the per-voxel agreement between a predicted tumor mask and a per-subject reference mask, measured in voxel space by Dice similarity coefficient, 95th-percentile Hausdorff distance (HD_{95}), and absolute volumetric error ($|VE|$), optionally per BraTS compartment (whole tumor WT, tumor core TC, enhancing tumor ET); reported in Table 3. **(ii) Anatomical localization** is the assignment of a tumor (or of each of its compartments NCR/ED/ET) to one or more anatomical regions of the patient’s own brain (here the 172 `wmparc` regions from the `SynthSeg/wmparc` parcellation produced in native subject space by `recon-all-clinical`, with no atlas warping) yielding a discrete region-set or laterality label rather than a similarity score. **(iii) Anatomical profiling** is the per-subject quantitative characterization of the overlap between (i) and (ii): the volume (or the per-voxel sigmoid probability mass) of every (compartment, region) pair, returned as a matrix or as a vector of relative regional tumor burdens. The Hit-Plot variants are specific instances of anatomical profiling: the volumetric Hit-Plot, the probabilistic Hit-Plot, the cohort-level functional-anatomy summary over pre-defined `wmparc` label groups, and the cortical-surface / tract-proximity / Hit-Plot-to-outcome variants named as future work in the Limitations subsection of the Discussion. Where the older term *anatomical fingerprint* is retained for continuity with the neuro-oncology literature [28], it is used as a synonym of *anatomical profiling* in sense (iii). A claim about (i) is therefore a claim about voxel-level agreement, a claim about (ii) is a claim about which brain regions are involved in a single subject, and a claim about (iii) is a quantitative, multi-region claim.

Evaluation framework We apply a single evaluation framework to every subject reported in this work (the $n = 50$ UCSF-PDGM evaluation cohort and the LUMIERE longitudinal subject Patient-048, Figure 6), built from three components that are shared across datasets and fixed at the start of the analysis. **(a) One segmentation engine.** The MONAI Bundle BraTS three-compartment SegResNet (Methods § *Tumor segmentation*) runs on every reported subject behind the same thin Python adapter. Neither dataset has a dataset-specific segmenter, and the LUMIERE-team HD-GLIO-AUTO and DeepBraTumIA masks are kept only as comparators against the unified engine (Table 5). The dataset-provided UCSF-PDGM masks are retained as a per-subject

reference for the segmentation-accuracy reporting in Table 3. **(b) One metric set.** Per-compartment (WT, TC, ET) Dice, HD_{95} , $|VE|$, sensitivity, and specificity for segmentation accuracy in sense (i) of the Terminology paragraph above, complemented by the per-(subject, region, compartment) Hit-Plot cell value for the spatial-pattern agreement claim in sense (iii), with concordance correlation coefficient (CCC) and Bland-Altman limits as the cohort-level summary. Both metric families are reported on the same $n = 50$ cohort, against the same reference masks, and computed by the same producer code paths. No dataset-specific metric variant is introduced. **(c) One per-subject report.** Every subject writes a fixed, schema-versioned set of artefacts in the same on-disk layout: a multi-class tumor label map and matching JSON sidecar (engine, device, wall-clock, channel order), a per-compartment metric set, the per-subject `wmparc` parcellation plus its sidecar, and the per-source Hit-Plot CSVs. The cohort-level tables and figures (Table 3, Table 4, and the Hit-Plot agreement panel) are pure aggregators of these per-subject artefacts. The pre-registered visual outlier audit (Methods § *Per-subject outlier audit on Table 3*) operates on the same per-subject sidecars. Adding a fourth dataset, or replacing the engine, therefore reduces to (re-)running the producer chain. The unified framework is what makes the cross-cell agreement summary in Table 3 and the Hit-Plot agreement panel directly comparable across subjects and across compartments without per-dataset caveats.

Per-subject outlier audit on Table 3 To audit segmentation accuracy at the per-subject (rather than cohort-aggregated) level, we apply a pre-registered two-stage outlier audit to the per-subject metric sidecars consumed by Table 3. (1) *Algorithmic flagging.* A subject is flagged if any compartment (WT, TC, ET) falls below the BraTS-2018 strict operating-point band ($WT \geq 0.85$, $TC \geq 0.75$, $ET \geq 0.70$ Dice) or exceeds $HD_{95} > 20$ mm on any compartment. (2) *Visual adjudication.* Every flagged subject is rendered as a canonical-RAS T1c/reference/DL panel at the lesion centroid (with worst-case singletons rendered standalone) and reviewed against three pre-registered hard-failure rules: lobar miss, contralateral hallucination, and artefact-dominated mask. A flagged subject is excluded from Table 3 only if at least one hard-failure rule is met. The full audit is reproducible from the per-subject sidecars. The resulting adjudication is reported in the corresponding Discussion paragraph and, for the worst-case boundary-disagreement exemplar (sub-0396), in Limitations item (vi).

The UCSF-PDGM dataset The UCSF-PDGM dataset consists of $n=501$ adult patient neuroimages with histopathologically confirmed WHO grade 2-4 diffuse gliomas [65]. The subjects underwent preoperative MRI examination using a 3T (Discovery 750 GE) MR scanner with a dedicated eight-channel head coil, with acquisitions according to a standardized brain tumor MR imaging protocol: T1-weighted precontrast, T2-weighted, T2-weighted FLAIR, T1-weighted postgadolinium, as well as susceptibility-weighted imaging and advanced diffusion and perfusion imaging techniques [22]; see Figure 9 (APPENDIX 1) for an example study. In the UCSF-PDGM dataset, all image data were manually reviewed for completeness and quality by a panel of reviewers with varying years of experience. All procedures, including imaging, initial tumor resection, and tumor genetic testing were performed at a single UCSF medical center between 2015 and 2021. For details regarding the multiparametric MRI acquisition parameters and the genetic biomarker testing with assessment of isocitrate dehydrogenase (IDH) mutations by genetic sequencing and PCR assay for measuring O⁶-methylguanine DNA methyltransferase (MGMT) promotor methylation status, see [19, 22, 21, 20]. The UCSF-PDGM team could not provide original DICOM data; the released “skull-stripped”, deidentified, and pre-processed image files

were prepared to facilitate the type of research that this dataset was intended for, where the channels (T1, T1c, T2, FLAIR, tumor-segmentation) are co-registered to a common matrix size of $240 \times 240 \times 155$ with isotropic 1 mm^3 voxels and data type `float64`. We use these released NIfTI volumes as provided, without additional inter-sequence coregistration, and limit preprocessing to visual quality control of alignment.

Data collection was performed in accordance with relevant guidelines and regulations and the UCSF-PDGM study was approved by the University of California San Francisco, institutional review board with a waiver for consent [22, 21]. The data, for which each patient has skull-stripped coregistered 3D images across the above-mentioned MRI sequences, as well as multicompartment tumor segmentations was made available via The Cancer Imaging Archive (TCIA) hosted at the University of Arkansas for Medical Sciences [21, 26]. Regarding image pre-processing and segmentation, each sequence was registered and resampled to the $1 \times 1 \times 1 \text{ mm}^3$ isotropic resolution space defined by the 3D FLAIR image. For image registration, an automated nonlinear registration based on the Advanced Normalization Tools (ANTs) [7] was used with previously published parameters [18]. Resampled coregistered images were then skull-stripped using Brain Mask (https://github.com/ecalabr/brain_mask), a publicly available 3D deep convolutional neural network-based segmentation and brain masking tool for multi-modal brain imaging, cf. [18].

The corresponding tumor ground truth that UCSF-PDGM provides has three tumor compartments: (i) the enhancing tumor, **ET** (mask value=4); (ii) the central non-enhancing/necrotic tumor, **NCR** (=1); and (iii) the surrounding FLAIR abnormality, including non-enhancing tumor and related edema, **ED** (=2). These regions were extracted using an automatic segmentation model followed by manual correction and approval by a team of experienced radiologists. More specifically, the segmentation of multicompartment tumors in the UCSF-PDGM dataset was carried out as a part of the 2021 BraTS challenge, following previously established methods reported in Baid et al. [8]. Initially, the image data was subjected to automated segmentation using a collective model made up of algorithms from prior BraTS challenges. Following this, the images were manually corrected by a team of annotators with different levels of experience. One of two neuroradiologists with over 15 years of professional experience approved the corrections. This process ensures that the ground truth is reasonably accurate and reliable.

Five subjects (Table 2) are retained as illustrative detail cases for the per-subject figures in the main text and appendices. They are not the $n = 50$ evaluation cohort of Table 1. For historical context, those five were originally chosen from the UCSF-PDGM cohort according to: (i) Diagnosis of glioblastoma, Isocitrate dehydrogenase (IDH) wildtype; and the diversifying criteria (ii) The sample should express variation in sex, age, tumor size, tumor location, and image quality.

The evaluation cohort is a leak-safe, clinically stratified sample of $n = 50$ GBM (IDH-wildtype, WHO grade 4) subjects from UCSF-PDGM v5. The draw is reproducible (fixed random seed, stratified by Sex $\times$ OS-tertile with cut points 213 and 484 days of overall survival), conditioned on a known overall-survival outcome and on the absence of biopsy prior to baseline imaging. Subjects belonging to the BraTS21 segmentation *Training* cohort are excluded to prevent training-data leakage into our evaluation. The full subject listing is pinned in a cohort manifest (cited in *Code and data availability*). Per-subject metadata are summarised in Table 1.

Field	Value
Subjects (n)	50
Sex	F: 20, M: 30
Age at MRI (years)	median 64; mean 62.1; range [29, 83]
Overall survival (days)	median 318; range [6, 1449]
OS tertile (short / mid / long)	short: 17, mid: 17, long: 16
MGMT status	indeterminate: 1, negative: 15, positive: 34
Extent of resection (EOR)	GTR: 31, STR: 13, biopsy: 6
Vital status (1=dead, 0=alive)	0: 19, 1: 31
BraTS21 segmentation cohort	Validation: 10, unused: 40

Table 1: **Cohort summary** ($n = 50$, UCSF-PDGM v5). Derived from the public UCSF-PDGM v5 metadata. OS = overall survival in days; EOR = extent of resection (GTR = gross total resection, STR = subtotal resection). The BraTS21 segmentation cohort row confirms zero subjects from BraTS21 *Training*, i.e. no leakage into our evaluation set.

ID	Sex	Age at MRI	MGMT status	MGMT index	Status	OS	EOR
0020	M	49	negative	0	Alive	551	GTR
0022	F	61	positive	14	Dead	533	STR
0039	M	53	positive	1	Alive	134	GTR
0066	F	38	unknown	-	Alive	1913	biopsy
0085	M	77	negative	0	Alive	232	GTR

Table 2: **Characteristics of five illustrative UCSF-PDGM subjects used for per-subject figures and Hit-Plot detail (not the $n = 50$ evaluation cohort of Table 1).** The dataset-provided multicompartiment tumor masks for these cases serve as *reference masks* for qualitative panels only. Cohort-level segmentation accuracy is reported separately on the $n = 50$ evaluation set (Table 3). ID = Subject number according to the UCSF-PDGM repository. Age is given in years at the time of MR imaging. MGMT = O6-methylguanine-DNA methyltransferase status, a clinical interpretation of the MGMT index using an in-house method developed by UCSF clinical labs where numeric values 0-17 indicate the number of promoter methylation sites. Status = Survival status of the patient at the last clinical follow-up. OS = Overall survival in days from the initial diagnosis to the last clinical follow-up. EOR = Extent of resection determined by review of operative reports and immediate postoperative imaging. GTR = Gross total resection. STR = Subtotal resection. All subjects had WHO CNS Grade 4 tumor and a final pathologic diagnosis of Glioblastoma, IDH-wildtype (WHO 2021).

To illustrate the input data (co-registered mpMRI recordings and tumor segmentation) to our pipeline, Figure 10 (APPENDIX 1) depicts a slice from the multiparametric MRI examination and corresponding color-coded tumor segmentation (used as a reference mask) in one of the selected subjects (ID 0020) from the UCSF-PDGM dataset.

The LUMIERE dataset

The LUMIERE dataset [88] represents the release of a unique longitudinal glioblastoma (GBM) MRI dataset from a single center, the University Hospital of Bern, Switzerland, aiming to enhance treatment response evaluation. Unlike many publicly available datasets that mainly feature pre-operative MRI or limited follow-up images, this dataset offers a comprehensive longitudinal view with expert ratings based on neuro-oncology

response assessment criteria (RANO) [25]. It provides additional pathology details for a patient subset, including O6-methylguanine-DNA methyltransferase (MGMT) promoter methylation and isocitrate dehydrogenase 1 (IDH1) status and overall survival (OS) times. The dataset, encompassing multiparametric MRI recordings (T1, T1-Gad, T2, and FLAIR) and state-of-the-art automated segmentation tools, comprises 91 GBM patients’ data with 638 study dates and 2487 images.

As an additional set of input data to our case-based feasibility study, we selected one subject (Patient-048, also depicted in [88]), a female, 52 yrs at surgery, 58 weeks survival time, IDH1 neg, and with MGMT not methylated. This patient had a total of six mpMRI examinations on different Siemens 1.5 T scanners over a period of 49 weeks from the primary examination (day 0), cf. Figure 11 (APPENDIX 1). Due to clinical and MR facility constraints, the four MRI sequences (T1, T1C, T2, FLAIR) were collected with different spatial resolutions, slice counts, timing parameters (TR, TE, TI, FA), and scan options at the different examinations (for details, see the `LUMIERE-MRinfo.csv` in <https://figshare.com/s/f3f5429e9e9275ad279d>). The channel images were converted from DICOM and skull-stripped using HD-BET [48]. They were further registered to the reference image (that differed by case), with orientation matching the MNI atlas.

We use all six examinations of Patient-048 (week-000-1, week-000-2, week-013, week-023, week-045, week-049) under the unified evaluation framework. A cohort manifest (cited in *Code and data availability*) records, per timepoint, the LUMIERE archive identifier, the anonymised `days_since_baseline` (0, 3, 91, 161, 315, 343), the RANO label assigned by the LUMIERE consortium (Pre-Op, Post-Op, SD, CR, PD, PD), and the per-timepoint mpMRI channel inventory. The two baseline scans (week-000-1, week-000-2) carry day-resolution from the LUMIERE archive (0 and 3 days, respectively). From week-013 onward the LUMIERE release reports timepoints as floor-weeks [88], so we report `days_since_baseline` for those four timepoints as $\text{week} \times 7$ with a ± 3 day uncertainty.

The four MR channels at every timepoint are staged in a clean BIDS-like layout, together with two LUMIERE-team reference segmentations used here as comparators only: HD-GLIO-AUTO (two-compartment, ED + NCR and ET + TC) and the native CT1-grid DeepBraTumIA mask (three-compartment). The native CT1-grid DeepBraTumIA variant is chosen over the SRI24-atlas variant that ships in the same archive because the former lives in the same physical space as the staged CT1 channel and therefore warps cleanly through the per-timepoint rigid transform (the atlas variant is in SRI24 space and would need a separate nonlinear inverse warp to recover the patient’s own anatomy).

All six timepoints are then brought into a single subject reference space (the pre-operative T1 of week-000-1 in canonical RAS) by a deterministic ANTsPyX rigid-registration orchestrator. The transform is estimated once per non-reference timepoint on a chosen *moving channel* (T1 by default; CT1 at week-023 because that timepoint’s T1 is acquired as a 6 mm-thick axial stack and is a poor target for high-resolution rigid registration) and then applied (linear interpolation) to all four MR channels and (nearest-neighbour) to both comparator segmentations. A per-timepoint quality-control sidecar records the moving-channel choice, the warped-versus-fixed Pearson correlation, the cross-timepoint same-channel correlations on the reference grid, and the fraction of each warped legacy-segmentation label that falls inside the reference brain mask.

The same unified MONAI Bundle BraTS three-compartment SegResNet that serves as the canonical segmentation engine for the $n = 50$ UCSF-PDGM evaluation cohort (Methods § *Evaluation framework*) is then run on the registered $\{\text{T1, CT1, T2, FLAIR}\}$ stack at every timepoint. The bundle expects LPS inputs, so the producer round-trips $\text{RAS} \rightarrow \text{LPS} \rightarrow \text{RAS}$ around the inference call to keep all on-disk artefacts in a single

canonical orientation. The DL-vs-comparator agreement is reported in Table 5. A single-pair HD-GLIO-AUTO + recon-all-clinical day-0/week-49 view of this patient, rendered manually with the FreeSurfer 7.4.1 toolchain, is preserved below as Figure 7 for qualitative context.

Methods for anatomical tumor localization and profiling

To estimate the tumor locations, we used FreeSurfer version 8.2.0 [35] (<https://surfer.nmr.mgh.harvard.edu>), an open-source software suite for processing human brain MRI (<https://github.com/freesurfer/freesurfer>). More specifically, we used its recon-all-clinical pipeline [12, 13, 38, 47]. The pipeline is implemented as a tcsh (Tenex C Shell) script for processing clinical scans of arbitrary orientation, resolution, and contrast. The script calls SynthSeg [12] for volumetric segmentations, which provide multiple labels covering a wide range of cortical and subcortical structures. More specifically, SynthSeg provides left (lh-) and right (rh-) hemispheric cortical *parcellations* (34 labels in lh and rh, respectively); left and right hemispheric *segmentations* (14 labels in lh and rh, respectively, including *accumbens*, *amygdala*, *caudate*, *hippocampus*, *lateral ventricles*, *pallidum*, and *thalamus*); in addition to *3rd ventricle*, *4th ventricle*, *brain-stem*, *csf*, and *background* - 101 labels altogether.

SynthSeg, with code and trained model, is open-sourced (<https://github.com/BBillot/SynthSeg>). It is based on a 3D UNet model developed by Billot et al. [11] and trained on brain scans with varied contrasts synthesized from pre-computed segmentations. Interestingly, the model is trained purely on synthetic scans produced by a generative model using a domain randomization (DR) physics-based strategy [93], where all the parameters of the generative model are fully randomized. SynthSeg only requires a set of anatomical label maps to be trained, no real recordings. These maps can be automatically obtained since the image data for training are directly generated from their ground truth and thus in perfect geometric correspondence. Note that segmentation ground truths were mainly obtained by running FreeSurfer on the T1 scans of each dataset in their study and undergoing a thorough visual quality control to ensure anatomical correctness; for details, see [12]. Their approach resulted in a model with remarkable generalization ability (... *augmentation beyond realism leads to improved generalization* [10]) that can be used for brain scans from a wide range of subject populations, including scans from young and healthy, aging, and diseased subjects and scans with white matter lesions, strokes, or tumors. Additionally, SynthSeg can handle scans with or without preprocessing, including bias field corruption, skull stripping, intensity normalization, and template registration [12]. It can provide reasonable segmentations even when there is a low signal-to-noise ratio. Therefore, it is feasible to use SynthSeg to identify anatomical structures in MRI sequences whose intensities are perturbed or corrupted by tumor growth.

Additional outputs from the recon-all-clinical pipeline are SynthSR [47], a synthesized, low-noise, super-resolution 1 mm³ isotropic MP-RAGE T1-w image, where lesions are inpainted i.e., the technique of filling in missing or damaged parts of an image; wmparc, with fine-grained white matter parcellations (172 labels including the *Brain-Stem* and *CSF*); aseg, a cleaned-up volume atlas of 40 subcortical regions using the reconstructed surfaces so that the cortical gray matter in the aseg volume follows the surface boundaries in the subject’s brain and other segmentations. By construction, all these are co-registered with the mpMRI recordings and the dataset-provided multicompartiment tumor reference mask. Our pipeline is illustrated in Figure 1.

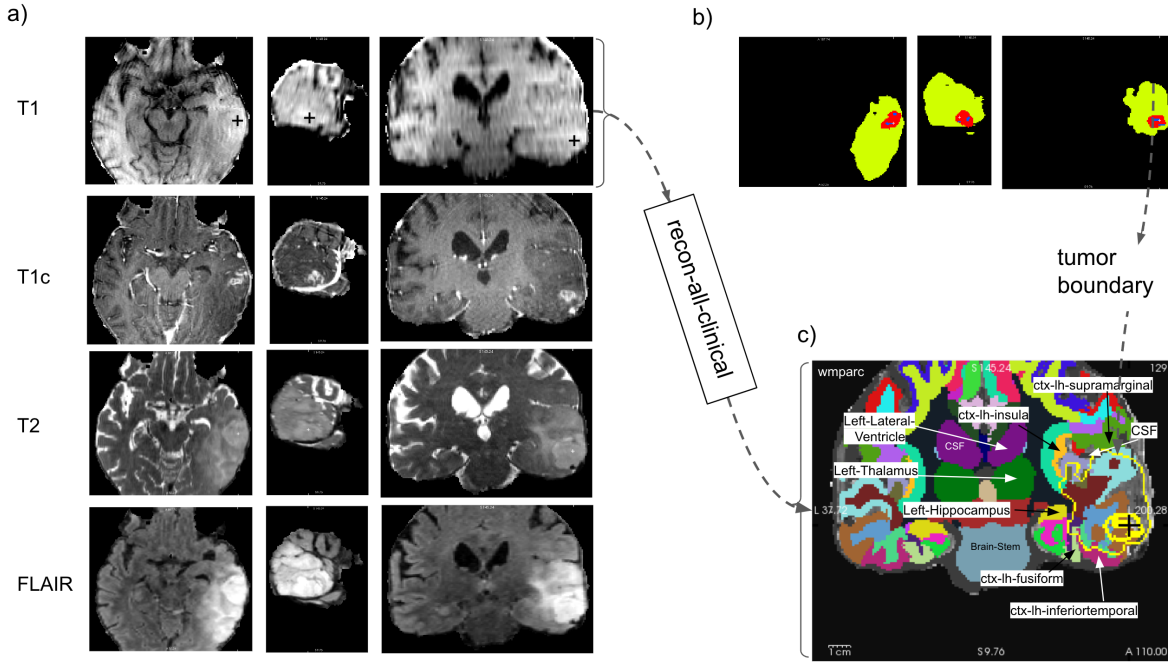


Figure 1: **The pipeline for anatomical tumor localization and profiling.** a) The skull-stripped and coregistered mpMRI recording (T1, T1c, T2, FLAIR) from UCSF-PDGM-0085. The black crosshair has a center $[i=182, j=129, k=71]$ (fixed for all images) located in the central part of the tumor (value 4 = ET); b) The color-coded, three-compartment tumor segmentation mask provided by UCSF-PDGM and used as a reference mask; c) wmparc segmentation. Visualization of normal and abnormal tissue in a coronal plane shown as annotated color-coded overlays, together with the tumor compartment boundaries, using the Freeview interactive visualization tool and the FreeSurferColorLUT lookup table.

For visual, interactive inspection of the mpMRI recordings and the corresponding segmentations, i.e., $\text{mpMRI} = [\text{T1}, \text{T1c}, \text{T2}, \text{FLAIR}, \text{tumor_segmentation}, \text{synthseg}]$, where $\text{mpMRI.shape} = (240, 240, 155, 5)$, and for producing figures, we used FreeView, the data viewer provided by FreeSurfer. For labeling SynthSeg and wmparc segmented regions (cf. Figure 1), we used the FreeSurferColorLUT.txt nomenclature for the anatomical regions in the left and right hemispheres. For anatomical profiling we compute per-subject Hit-Plots as combinatorial voxel counts between each BraTS compartment (NCR, ED, ET) and each non-empty wmparc label (172 labels including brain-stem and CSF). Overlap volumes are reported in mL and, where noted, as fractions of intracranial volume (ICV). For a cohort-level summary of where in the brain the leak-safe $n = 50$ UCSF-PDGM evaluation cohort sits anatomically, we bin every wmparc label into 11 anatomical lobes (frontal, cingulate, insula, parietal, temporal, occipital, sub-cortical deep, sub-cortical limbic, cerebellum, brainstem, other) using a canonical mapping, and aggregate the per-subject Hit-Plot CSVs by summing the per-(label, compartment) overlap volume within each lobe. Across the cohort, the whole-tumor compartment intersected the frontal lobe in 90 % of subjects (45/50), the parietal lobe in 90 % (45/50), the insula in 80 % (40/50), the sub-cortical deep nuclei (thalamus, basal ganglia, ventral DC) in 80 % (40/50), the temporal lobe in 70 % (35/50), the occipital lobe in 48 % (24/50), the sub-cortical limbic structures (hippocampus, amygdala) in 52 % (26/50), and infratentorial structures (cerebellum and brain-stem) in 20 % (10/50) and 22 % (11/50) respectively, consistent with the supratentorial-dominant incidence distribution of adult diffuse glioma reported by Larjavaara et al. [56]. A per-region anatomical-profile sunburst representation is readable at the single-subject level but aggregates poorly across the cohort. We therefore deliberately do not render the sunburst as a manuscript figure here. Instead, the Appendix re-

ports a compact functional-anatomy Hit-Plot summary that keeps the analysis descriptive: the same per-subject `wmparc` Hit-Plot CSVs are aggregated into pre-defined label groups (lobar footprint, motor-adjacent cortex, left language-adjacent cortex, and deep-midline / commissural structures), with prevalence of any overlap and median share of compartment volume reported for WT, TC, and ET, including exploratory OS-tertile panels. These groups are anatomical-label adjacency summaries, not tractography, individualized eloquence mapping, fMRI, organ-at-risk proximity, treatment-planning validation, or outcome prediction (Discussion § *Limitations and translational scope*, item (vii)).

Tumor segmentation: reference masks and the unified deep-learning engine

The canonical tumour-segmentation engine for both reported datasets is the MONAI Bundle BraTS three-compartment SegResNet, run behind a thin Python adapter that exposes a single $\{T1, CT1, T2, FLAIR\} \rightarrow \{NCR, ED, ET\}$ contract on a canonical-RAS grid. The three resulting compartments use the BraTS label scheme ($NCR = 1$, $ED = 2$, $ET = 4$) and are aggregated downstream into the BraTS sub-regions WT ($NCR \cup ED \cup ET$), TC ($NCR \cup ET$), and ET. The dataset-provided UCSF-PDGM masks are retained as a per-subject *reference* for the segmentation-accuracy reporting in Table 3, not as a substitute engine. For the LUMIERE longitudinal subject, two LUMIERE-team segmentations are kept as *comparators* only: HD-GLIO-AUTO (two-compartment) and DeepBraTumIA-native (three-compartment); their per-(timepoint, compartment) agreement against the unified engine is reported in Table 5.

All processing was performed in Python 3.11 using the pinned software stack declared in `pyproject.toml` and `environment.yml` (NumPy, Pandas, Matplotlib, Nibabel, MONAI, ANTsPyX, FreeSurfer 8.2.0). Deep-learning segmentation on the $n = 50$ evaluation cohort used the MONAI Bundle 0.4.8 BraTS-3 SegResNet backend on the stated hardware (Table 4). The repository and full stack are cited in *Code and data availability*.

3 Results

We first report cohort-level segmentation accuracy of the unified MONAI Bundle pipeline against the UCSF-PDGM reference masks across all $n = 50$ subjects (Table 3), and then illustrate end-to-end pipeline behaviour (segmentation, parcellation, anatomical neighbourhood, and Hit-Plot) on representative cases from the UCSF-PDGM cohort and on the LUMIERE longitudinal subject (Patient-048). The remaining three UCSF-PDGM cases are shown in the Appendix.

Quantitative segmentation accuracy on the $n = 50$ UCSF-PDGM cohort

The unified MONAI Bundle BraTS three-compartment SegResNet (Methods § *Tumor segmentation*) was run on all $n = 50$ subjects of the leak-safe UCSF-PDGM evaluation cohort (Table 1). Per-subject DL outputs were compared against the dataset-provided tumor masks using the operational definition of segmentation accuracy fixed in Methods § *Terminology*, sense (i): Dice similarity coefficient, 95th-percentile Hausdorff distance (HD_{95}), absolute volumetric error ($|VE|$), sensitivity and specificity, computed per BraTS compartment (whole tumor WT, tumor core TC, enhancing tumor ET). Cohort-level distributions are summarised as median followed by the [Q1, Q3] inter-quartile range in Table 3. Median Dice is 0.901, 0.941, 0.862 for WT, TC, ET respectively, i.e. above the BraTS-2018 strict operating-point band ($WT \geq 0.85$, $TC \geq 0.75$, $ET \geq 0.70$ Dice) on the cohort median. The pre-registered outlier audit on these per-subject metrics (Methods § *Per-subject outlier audit on Table 3*; Limitations item (vi)) flagged 16/50 subjects against the same strict band, and visual adjudication retained all sixteen.

Metric	WT	TC	ET
Dice	0.901 [0.850, 0.932]	0.941 [0.900, 0.963]	0.862 [0.793, 0.898]
HD_{95} (mm)	4.12 [2.45, 6.87]	2.24 [1.00, 4.30]	2.00 [1.41, 3.45]
$ VE $ (mm ³)	8170 [3534, 15288]	1385 [619, 2618]	2430 [1065, 5741]
Sensitivity	0.875 [0.784, 0.902]	0.927 [0.864, 0.955]	0.793 [0.729, 0.842]
Specificity	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]

Table 3: **Per-subject segmentation accuracy of the unified MONAI Bundle pipeline against the UCSF-PDGM reference masks** ($n = 50$). Each cell reports the cohort-level median followed by the [Q1, Q3] inter-quartile range across $n = 50$ subjects. WT, TC, ET refer to the BraTS subregions (whole tumor, tumor core, enhancing tumor). HD_{95} in millimetres, $|VE|$ (absolute volumetric error) in cubic millimetres. Per-voxel specificity is reported for completeness but saturates at 1.000 in every cell: the denominator $TN + FP$ is dominated by non-tumour brain voxels ($\sim 10^6$ background vs. $\sim 10^3$ false positives per subject), so even substantially over-segmenting predictions round to the same value and the metric is not discriminative for this task. Dice, HD_{95} , $|VE|$, and sensitivity carry the substantive comparison. The dataset-provided tumor masks are treated as a *reference* rather than ground truth (Limitations item (vi); the worst-case cross-segmenter boundary-disagreement exemplar sub-0396 is discussed there).

Per-pipeline-stage runtime on stated hardware

Wall-clock runtime for the two computational stages of the unified pipeline, measured on the same Apple Silicon host that processed the $n = 50$ evaluation cohort, is summarised in Table 4. The MONAI Bundle BraTS-3

SegResNet stage runs in ≈ 1.3 s per subject on the steady-state path (Metal Performance Shaders, MPS), with a one-time first-call latency of ≈ 22 s per process for model-weight loading and MPS-context initialisation. The FreeSurfer `recon-all-clinical` anatomical parcellation stage runs single-host, multi-threaded on CPU and is the dominant contributor to total cohort runtime. Per-subject elapsed time is captured as a structured field in every per-subject sidecar so the table is fully regenerable from the on-disk derivatives, with a per-subject audit CSV and a JSON companion that records the host fingerprint. The pinned software stack and repository are cited in *Code and data availability*.

Stage	Backend	Device	n	Wall-clock per subject (median [Q1, Q3])
DL segmentation (first-call latency)	MONAI Bundle 0.4.8 (BraTS-3 SegResNet)	MPS	1	22.01 s
DL segmentation (steady-state)	MONAI Bundle 0.4.8 (BraTS-3 SegResNet)	MPS	49	1.29 s [1.28 s, 1.35 s]
Anatomical parcellation	FreeSurfer 8.2.0 recon-all-clinical (wmparc)	CPU/10t	50	35:25 min [32:38 min, 41:08 min]

Table 4: **Per-pipeline-stage wall-clock runtime on stated hardware** ($n = 50$ **evaluation cohort**). Auto-generated from the per-subject elapsed-time fields in the on-disk derivatives. The DL segmentation stage is split into a single “first-call latency” row (cold-start cost paid once per process for MONAI Bundle weight loading and MPS-context initialisation) and a “steady-state” row (the remaining $n-1$ subjects). Cells report median followed by the [Q1, Q3] inter-quartile range, auto-formatted as seconds, MM:SS, or HH:MM:SS depending on magnitude. The n column reports cells where data are present out of the cohort total ($n_{\text{used}}/n_{\text{expected}}$ when different); a partial-cohort denominator on the parcellation row indicates that the long-running FreeSurfer fan-out was still in flight when this table was last regenerated. Re-running the runtime benchmark after the fan-out completes refreshes all cells in place.

Spatial-pattern agreement: Hit-Plot panel (DL vs UCSF-PDGM reference)

Spatial-pattern agreement between the unified DL pipeline and the dataset-provided reference masks was assessed on a per-(subject, anatomical region, BraTS compartment) basis using the Hit-Plot regional-tumor-burden representation (Methods § *Methods for anatomical tumor localization and profiling*). With the FreeSurfer `recon-all-clinical` parcellation acting as the binning grid, every cell encodes the percentage of one parcel occupied by one BraTS compartment, in one subject. Across all $n = 50$ subjects $\times$ 172 wmparc regions = 8 600 cells per compartment, the concordance correlation coefficient (CCC) between DL and reference is 0.974 (WT), 0.992 (TC), and 0.951 (ET); mean differences are -0.53 , -0.07 , and -0.24 percentage points of the underlying parcel volume, with 95% Bland-Altman limits of agreement $[-7.73, +6.67]$ (WT), $[-2.20, +2.06]$ (TC), and $[-3.47, +2.99]$ (ET) percentage points; mean absolute volumetric differences per cell are 80.9, 17.1, and 25.3 mm³, respectively. Figure 2 renders the per-compartment Bland-Altman scatter against the cell-mean.

Beyond the volumetric Hit-Plot reported here, the pipeline also computes a probabilistic Hit-Plot that propagates the segmenter’s per-class sigmoid probabilities into a per-cell Poisson-binomial mean and standard deviation. These per-subject outputs are produced by the released build chain, and a dedicated analysis of the propagated per-cell uncertainty is beyond the scope of the present feasibility report.

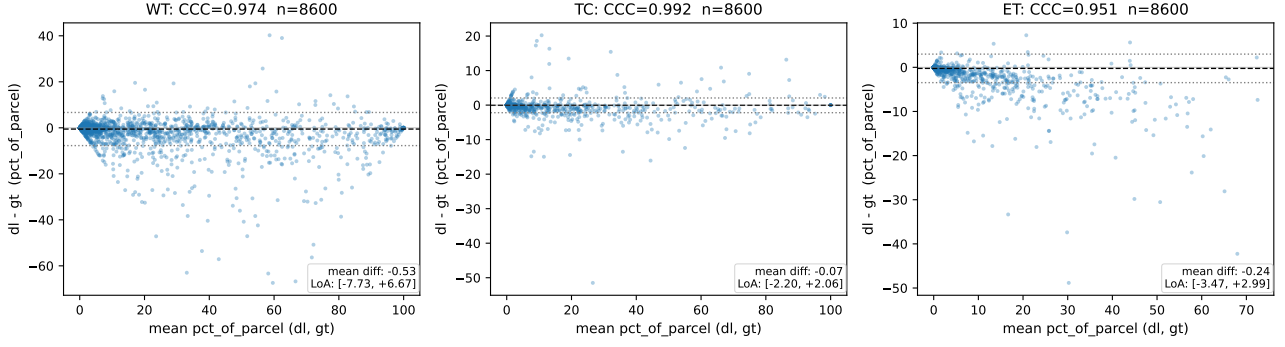


Figure 2: **Cell-level spatial-pattern agreement between the unified DL pipeline and the UCSF-PDGM reference masks** ($n = 50$ cohort, 172 wmparc regions, three BraTS compartments; 8 600 cells per compartment). Each panel is a Bland-Altman scatter of the per-(subject, region) Hit-Plot cell value (percentage of the parcel occupied by the BraTS compartment) for one compartment (WT, TC, ET). Solid horizontal line: cohort mean difference (-0.53 , -0.07 , -0.24 pp for WT, TC, ET); dashed lines: 95% limits of agreement (mean $\pm 1.96 \sigma_{\text{diff}}$; $[-7.73, +6.67]$, $[-2.20, +2.06]$, $[-3.47, +2.99]$ pp). Concordance correlation coefficient (CCC; 0.974, 0.992, 0.951) and mean absolute volumetric difference per cell ($|\Delta V|$; 80.9, 17.1, 25.3 mm^3) are reported in the panel headers. The companion audit CSV and machine-readable summary sidecar are archived alongside the figure (cf. *Code and data availability*).

Functional-anatomy summary of the Hit-Plot

To make the per-region Hit-Plot representation readable at the cohort level without claiming individualized functional mapping, we also aggregate the same $n = 50$ wmparc Hit-Plot CSVs into pre-defined anatomical-label groups (Appendix 4, Figure 16). The whole-tumor (WT) compartment intersects motor-adjacent labels in 44/50 subjects (88%), with a median 7.97% of WT volume falling in that group; left language-adjacent labels in 27/50 subjects (54%), median 0.79% of WT volume; and deep-midline / commissural labels in 40/50 subjects (80%), median 1.72% of WT volume. The same producer run on Hit-Plot-DL yields similar descriptive summaries (motor-adjacent WT 43/50, median 6.91%; left language-adjacent WT 27/50, median 0.71%; deep-midline / commissural WT 42/50, median 2.06%), consistent with the cell-level DL-vs-reference agreement in Figure 2. A bounded `mriTumorSynth` segmentation-backbone sensitivity run on the same $n = 50$ staged inputs gives the same qualitative functional-anatomy pattern (Appendix 5): motor-adjacent WT 43/50, median 6.27%; left language-adjacent WT 27/50, median 0.84%; and deep-midline / commissural WT 41/50, median 1.28%. These values are presented as an appendix-level anatomical readability check of the Hit-Plot variants, not as tractography, functional eloquence mapping, treatment-planning validation, or outcome prediction. The OS-tertile panels in Figure 16 demonstrate only that the same descriptive summaries can be stratified by available clinical metadata. They are not a test of regional-burden/outcome association.

Cases from the UCSF-PDGM dataset

We illustrate end-to-end pipeline behaviour on five *illustrative* UCSF-PDGM detail subjects (Methods, Table 2; not the $n = 50$ evaluation cohort). The pipeline is individually evaluated and visually assessed with a detailed, interactive inspection of the original recordings, the co-registered tumor masks (used as a reference),

and the normal tissue segmentation masks from `FreeSurfer`, including `SynthSeg`.

Figure 3 shows the segmentation results for a single subject (id=0020). The segmentations and tumor localization profiles (“Hit-Plots”) for this and subject (id=0022) are shown in Figures 4, 5, respectively; the corresponding figures for the three appendix subjects (id=0039, 0066, 0085) are Figures 12, 13, 14. The bar-chart bottom half of each per-subject figure (panel c of Figures 4, 5, 12, 13, 14) is computed from the `FreeSurfer 8.2.0 recon-all-clinical wmparc` parcellation. Bars are stacked NCR (red, label 1) + ED (green, label 2) + ET (yellow, label 4); their sum equals the whole-tumor volume in each region, and per-region totals (mL) are annotated to the right of each bar (NCR = Necroses, non-enhancing tumor; ED = FLAIR abnormality, edema compartment; ET = Enhancing tumor). These five detail subjects are leak-unsafe for our DL engine by construction (four of five sit in the BraTS21 Segmentation Training cohort and the fifth in Validation), so the per-subject Hit-Plot bars in the main text are reference-mask-based; the supplementary 5×1 DL-vs-reference panel of APPENDIX 3 (Figure 15) demonstrates per-subject DL-vs-reference agreement on five representative subjects from the leak-safe $n = 50$ evaluation cohort.

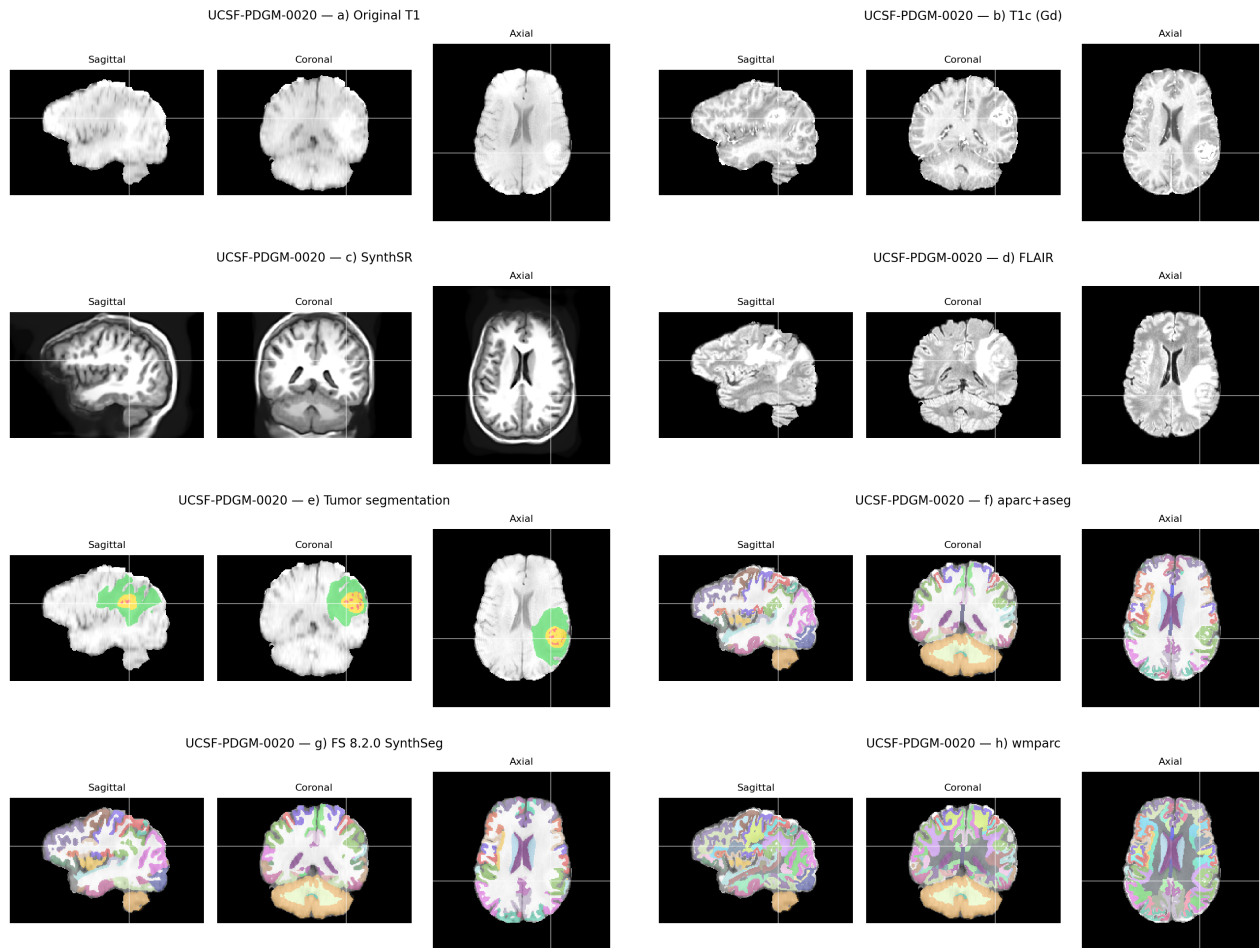


Figure 3: Segmentation of tumor (provided by the UCSF-PDGM team) and probable underlying anatomical regions in study UCSF-PDGM-0020. All eight panels are produced end-to-end with the unified HPGS pipeline: `mri_synthsr` produces the super-resolution T1; `recon-all-clinical.sh` (FreeSurfer 8.2.0) produces `aparc+aseg` and `wmparc`; and `mri_synthseg --robust --parc` produces the SynthSeg parcellation. The FS outputs are resampled back to the native mpMRI grid with `mri_convert -rl` for voxel-aligned display. All panels are displayed in the conventional *radiological* orientation on canonical-RAS data (image-left = patient-right for the axial and coronal planes; anterior-left for the sagittal plane; superior up for the axial and sagittal planes). a) Original T1. The black crosshair is fixed at $[i=159, j=147, k=95]$ in all panels and is centred in the tumor; b) Original Gadolinium-contrast enhanced T1-weighted image (T1c); c) SynthSR [47] super-resolution 1 mm MPRAGE synthesised from the original T1; d) Original FLAIR image; e) Three-compartment tumor segmentation mask (NCR=1, ED=2, ET=4) overlaid on the bias-corrected T1 used by HPGS; f) `aparc+aseg` from `recon-all-clinical.sh`; g) FreeSurfer 8.2.0 SynthSeg parcellation from the same subject's T1; h) `wmparc` from `recon-all-clinical.sh`. Color-coded label overlays follow the `FreeSurferColorLUT` lookup table.

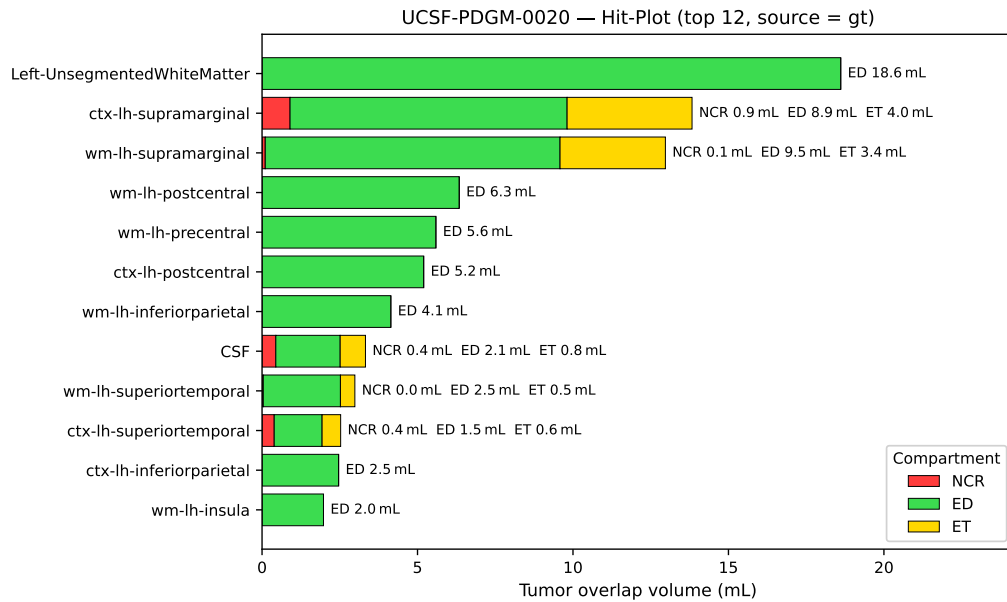
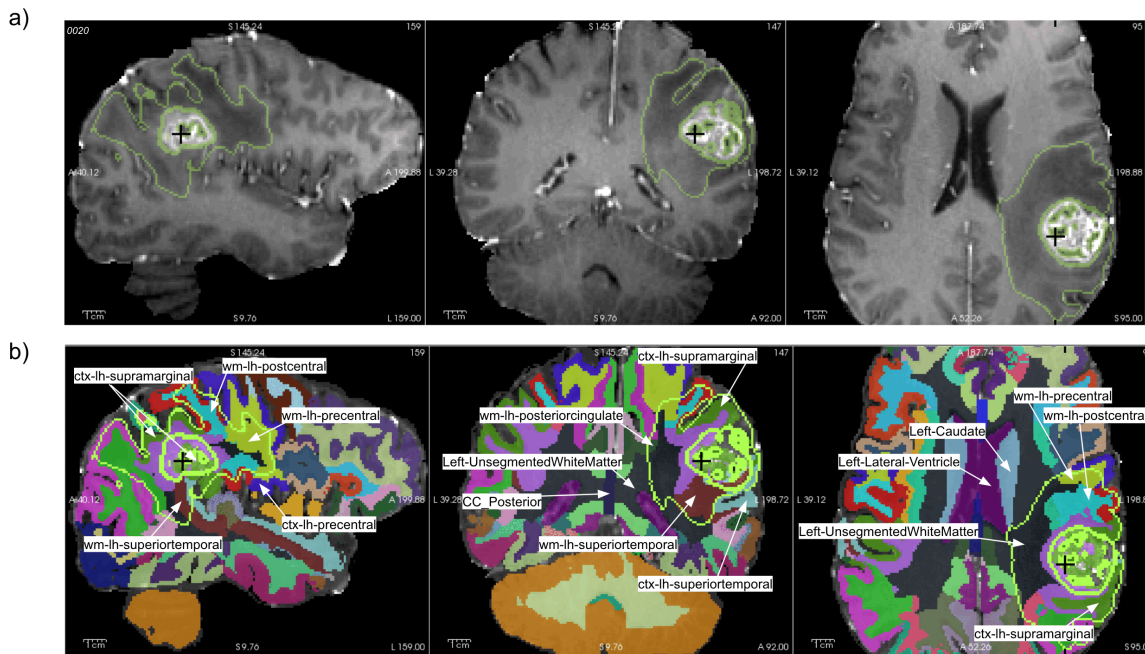


Figure 4: **Tumor location and anatomical neighborhoods from the study UCSF-PDGM-0020.**

- a) The original Gadolinium-contrast enhanced T1-weighted image (T1c) with an overlay of tumor region boundaries.
- b) The `wmparc` segmentation with an overlay of tumor compartment boundaries and annotation of selected anatomical structures and regions in the tumor surroundings. The black crosshair center is located at $[i=159, j=147, k=95]$. Visualization of normal and abnormal tissue in the sagittal, coronal, and axial planes are made with `Freeview` and color-coded labels according to the `FreeSurferColorLUT` lookup table.
- c) Tumor compartment “Hit-Plot” for the top-12 `wmparc` parcels of this subject.

One case from the LUMIERE dataset with longitudinal examinations

We process all six mpMRI examinations of Patient-048 in the LUMIERE collection (cf. Figure 11) under the unified evaluation framework: per-timepoint rigid coregistration into a single pre-operative reference space (Methods § *The LUMIERE dataset*), then per-timepoint segmentation by the unified MONAI Bundle BraTS three-compartment SegResNet on the registered {T1, CT1, T2, FLAIR} stack. The resulting BraTS three-compartment trajectory (NCR, ED, ET, plus the derived $TC=NCR+ET$ and $WT=NCR+ED+ET$) is shown in Figure 6, on a continuous day-resolved time axis with the LUMIERE-anonymisation disclosure propagated into the caption. Quantitatively, the trajectory recovers the expected RANO-aligned dynamics for this subject: a maximal-tumour pre-operative scan ($WT = 140.1$ mL, $ET = 24.0$ mL at day 0), a post-operative drop in enhancing tumour ($ET = 3.7$ mL at day 3, a $\sim 85\%$ reduction), a mid-treatment trough at week-013 (SD; $ET = 0.7$ mL, $WT = 31.5$ mL) followed by a CR plateau at week-023 ($ET = 1.0$ mL), and a clear recurrence by week-049 (PD; $ET = 22.7$ mL, $WT = 100.0$ mL).

Agreement between the unified-engine segmentation and the two LUMIERE-team reference segmentations (HD-GLIO-AUTO and the native CT1-grid DeepBraTumIA mask, both warped through the same per-timepoint rigid transform) is summarised compartment-by-compartment in Table 5. Two qualitative patterns are worth noting. (a) Whole-tumour (WT) and tumour-core (TC) agreement against DeepBraTumIA is consistently high at the high-burden timepoints (WT Dice 0.85 pre-op, 0.86 post-op, 0.90 at week-049; TC Dice 0.95, 0.72, 0.92 at the same three timepoints) and degrades, as expected, at the low-burden CR/SD mid-trajectory where the absolute number of tumour voxels is small and the Dice denominator becomes noise-sensitive. (b) Enhancing-tumour (ET) Dice tracks the absolute ET volume rather than segmenter identity: it is high when ET is genuinely present (Dice ≥ 0.77 at the pre-op and PD scans for both comparators) and collapses when ET essentially vanishes (Dice ≤ 0.13 at week-000-2 / week-013 / week-023, where the ET volume is below ~ 4 mL across all three segmenters and HD-GLIO-AUTO in particular reports near-zero ET). This is a volume-floor artefact of the Dice metric, not a discrepancy in the underlying clinical reading: all three engines agree, on these three timepoints, that there is essentially no enhancing tumour.

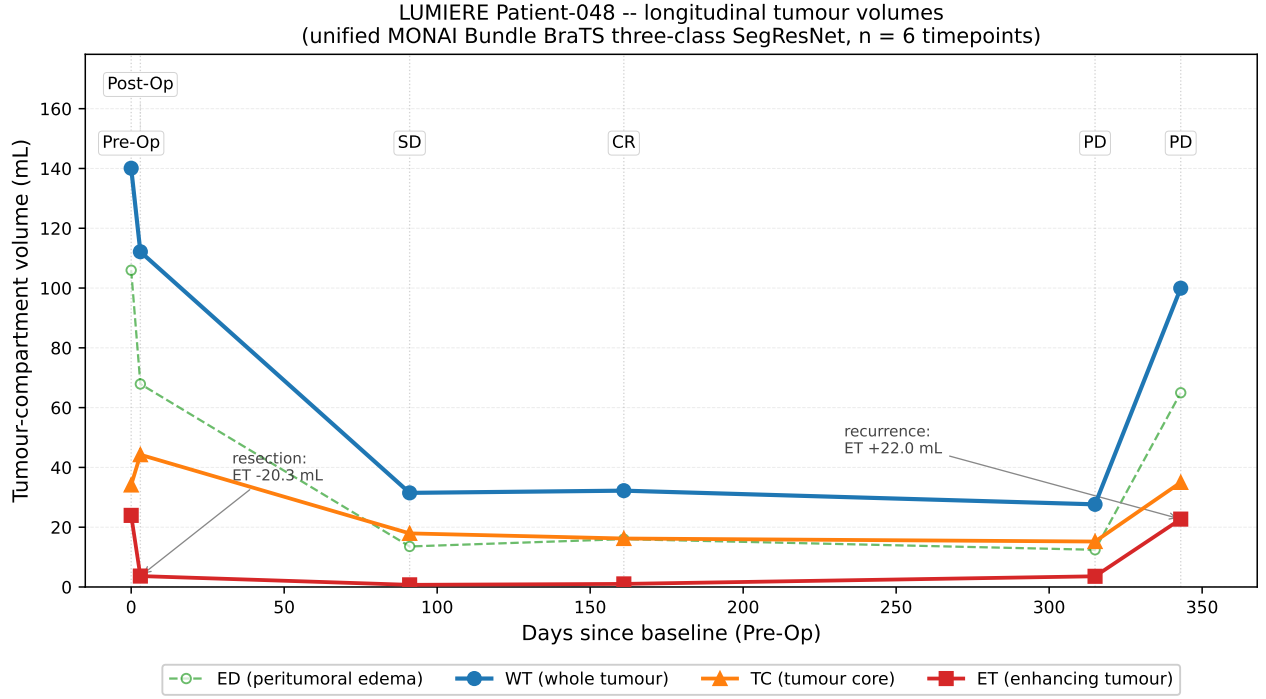


Figure 6: **LUMIERE Patient-048 – longitudinal tumour volumes from the unified MONAI Bundle BraTS three-compartment SegResNet ($n = 6$ timepoints).** Per-compartment volume (mL) on a continuous day-resolved time axis: enhancing tumour (ET, red), necrotic core (NCR, blue), tumour core (TC = NCR + ET, dashed), edema (ED, green), and whole tumour (WT = NCR + ED + ET, black). RANO labels (Pre-Op, Post-Op, SD, CR, PD, PD) and selected clinical event annotations (post-operative ET drop; recurrence trough/peak) are shown above the trajectories. *LUMIERE anonymisation disclosure:* the two baseline scans (week-000-1, week-000-2) carry day-resolution from the LUMIERE archive (0 and 3 days, respectively); from week-013 onward LUMIERE reports timepoints as floor-weeks [88], so days for week-013 / -023 / -045 / -049 are computed as $\text{week} \times 7$ and carry a ± 3 day uncertainty.

Timepoint (RANO, day)	HD-GLIO-AUTO		DeepBraTumIA (native CT1)			ET vol. (mL)
	WT	ET	WT	TC	ET	
week-000-1 (Pre-Op, d 0)	0.80	0.45	0.85	0.95	0.90	24.0
week-000-2 (Post-Op, d 3)	0.73	0.00	0.86	0.72	0.11	3.7
week-013 (SD, d 91)	0.17	0.03	0.70	0.76	0.11	0.7
week-023 (CR, d 161)	0.33	0.27	0.76	0.76	0.13	1.0
week-045 (PD, d 315)	0.40	0.37	0.78	0.85	0.61	3.6
week-049 (PD, d 343)	0.81	0.77	0.90	0.92	0.88	22.7

Table 5: **LUMIERE Patient-048: agreement between the unified MONAI Bundle BraTS three-compartment SegResNet and the two LUMIERE-team comparator segmentations, per timepoint and per compartment (Dice).** HD-GLIO-AUTO is two-compartment (WT, ET only); DeepBraTumIA (native CT1-grid variant) is three-compartment (WT, TC, ET). The right-hand column reports the unified-engine ET volume in mL at each timepoint as a context for reading the ET-Dice column: ET-Dice tracks ET burden rather than segmenter identity ($\text{Dice} \geq 0.77$ when ET volume > 20 mL, $\text{Dice} \leq 0.13$ when ET volume < 4 mL across all three engines), a known volume-floor artefact of Dice on small reference masks.

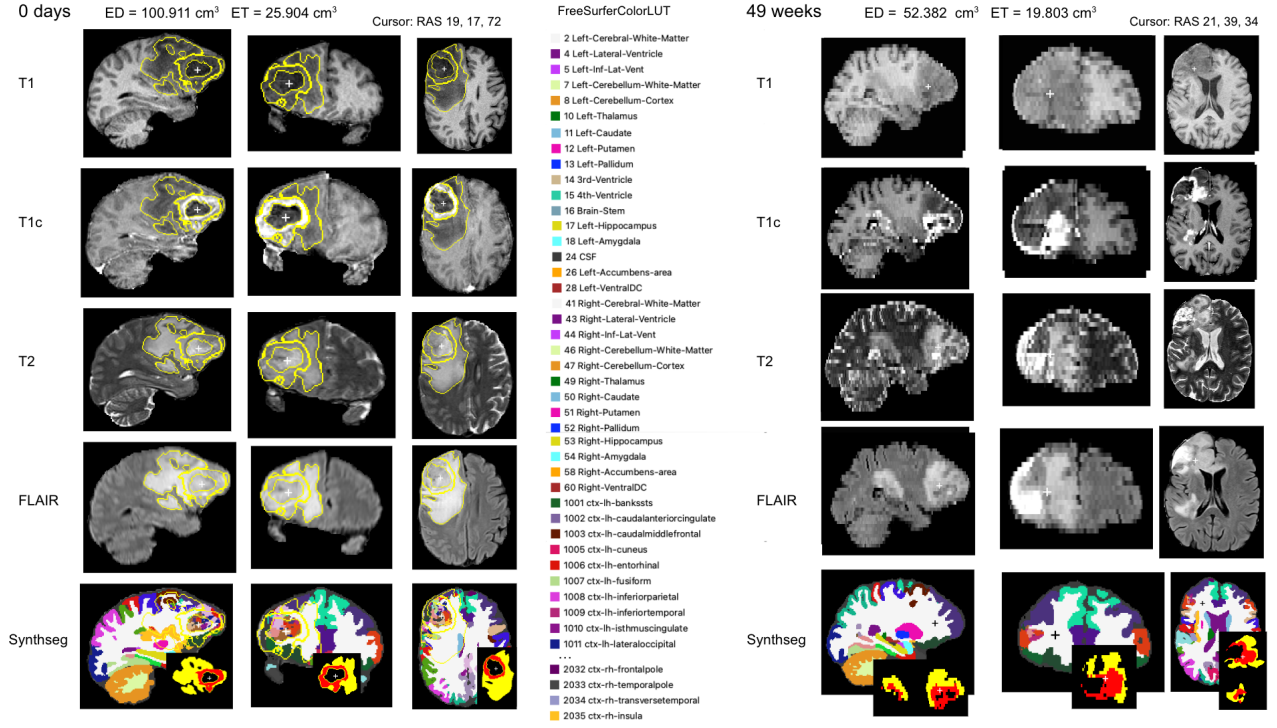


Figure 7: **Segmentations of the tumor located in the right frontal lobe and probable underlying anatomical regions in the longitudinal examinations of Patient-048.** For this case the `FreeSurfer 7.4.1 recon-all-clinical` pipeline was applied to the T1-weighted acquisitions in in each of the longitudinal examinations of the patient (cf. Figure 11). Data from the first visit (day 0) is shown in the left panel, data from the last visit (week 49) is shown in the right panel. Visualization of normal and abnormal tissue in the sagittal, coronal, and axial planes are provided by color-coded overlays using `Freeview` and the `FreeSurferColorLUT` lookup table. The two-compartment tumor segmentations, obtained from HD-GLIO-AUTO, were provided by the LUMIERE team. *Provenance:* the HD-GLIO-AUTO tumor segmentations are courtesy of the LUMIERE consortium [88], and the underlying longitudinal mpMRI examinations are part of the publicly released LUMIERE dataset (<https://github.com/ysuter/gbm-data-longitudinal>). The per-subject anatomical overlays in this figure were rendered manually in `Freeview` on `FreeSurfer 7.4.1 recon-all-clinical` outputs and provide qualitative context only; the quantitative longitudinal counterpart, computed end-to-end with the unified pipeline (DL segmentation $\cap$ `FreeSurfer 8.2.0 recon-all-clinical wmparc`), is the longitudinal Hit-Plot of Figure 8.

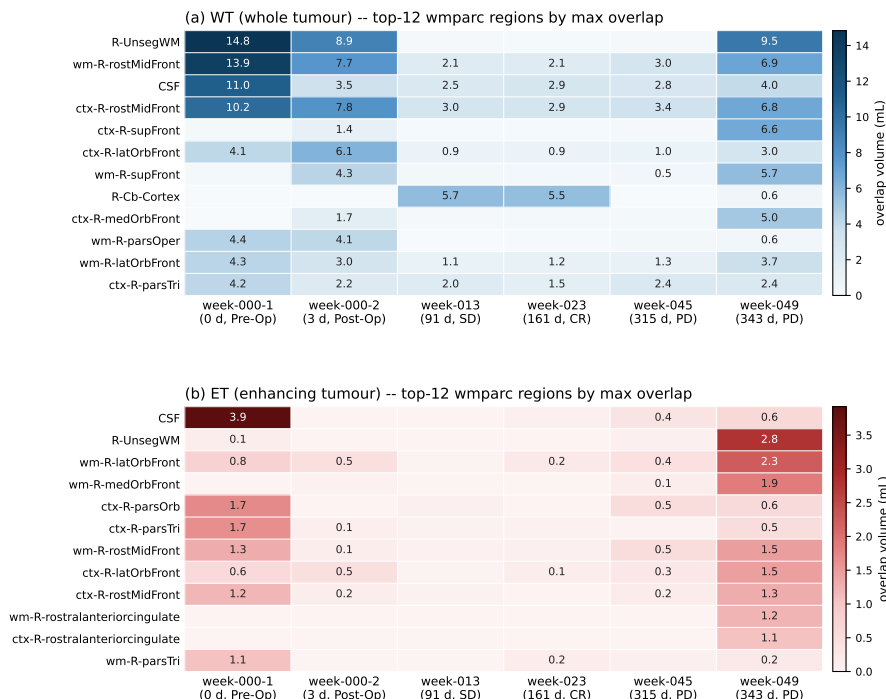


Figure 8: LUMIERE Patient-048: longitudinal Hit-Plot across the six examinations. The anatomy-aware Hit-Plot representation that powers the cohort-level DL-vs-GT agreement panel (Figure 2) is here applied across the Patient-048 longitudinal arc, closing the loop opened by the volume trajectories of Figure 6: the unified MONAI Bundle BraTS three-compartment SegResNet (Methods § *Evaluation framework*) is intersected with the FreeSurfer 8.2.0 recon-all-clinical wmparc parcellation on the registered RAS grid of each timepoint. *Top panel* (a) shows the top-12 wmparc regions ranked by maximum WT (whole-tumour) overlap volume across the six visits; cell colour and annotation give the per-cell overlap volume in mL. The Pre-Op (week-000-1, day 0) burden is dominated by right-hemispheric frontal white matter and cortex (wm-rh-rostralmiddlefrontal, ctx-rh-rostralmiddlefrontal, and the FreeSurfer Right-UnsegmentedWhiteMatter catch-all that aggregates tumour-cavity-adjacent white matter), drops sharply at the immediate Post-Op visit (week-000-2, day 3), stabilises through the SD/CR window (week-013, week-023; days 91/161), and recovers in the same anatomical territory at the terminal PD scan (week-049, day 343). *Bottom panel* (b) shows the top-12 wmparc regions for the ET (enhancing-tumour) compartment: the SD and CR visits show near-zero ET across the board (consistent with the radiologically reported clinical CR), and the PD recurrence at week-049 is anatomically colocalised with the original tumour bed, confirming that the right-frontal compartmentalisation is preserved end-to-end through resection → chemoradiation → recurrence. The Unknown (FreeSurfer label 0, the catch-all for tumour-cavity/mass-effect-distorted territory that did not receive a specific anatomical assignment) is omitted from both panels because it carries no anatomical interpretation. *Anonymisation disclosure:* the LUMIERE consortium [88] releases visit identifiers as floor-week stamps for re-identification protection; the days-since-baseline labels on the x-axis use the manifest-canonical (0, 3, 91, 161, 315, 343) sequence.

4 Discussion

In this feasibility study, we have demonstrated detailed quantitative and automated anatomical profiling from mpMRI in a leak-safe sample of $n = 50$ adult subjects with IDH-wildtype WHO grade 4 glioblastoma drawn from UCSF-PDGM v5 (Methods § *Evaluation framework*; Table 1), complemented by the LUMIERE longitudinal subject Patient-048 under the same unified evaluation framework. This is done to assess macroscopic tumor proliferation and how the tumor alters and intersects with brain architecture. The pipeline involves combined 3D segmentation of (i) brain tumor compartments and (ii) the possible underlying and adjacent brain structures for each tumor component. A major tool in this profiling was the use of `SynthSeg`, a deep learning tool for the segmentation of brain scans of any contrast and resolution.

Our contribution differs in several ways from the methodological approach implemented in `Raidionics` [15]. For the non-local openly available data, the most significant differences are: (i) We use tumor segmentations that are pre-computed. In case of UCSF-PDGM, the dataset-provided masks (treated as a per-subject *reference* rather than ground truth, see Methods § *Evaluation framework* and Limitations item (vi)), depending jointly on the multiple channels in the mpMRI recordings (Figures 10 and 11). This provides three biologically different tumor compartments: ET, NCR, and ED. For the LUMIERE longitudinal subject (Patient-048) we segment all six examinations with the unified MONAI Bundle BraTS three-compartment SegResNet (Methods § *The LUMIERE dataset*; Figure 6) and treat the LUMIERE-team HD-GLIO-AUTO and native CT1-grid DeepBraTumIA masks as comparators only (Table 5). `Raidionics` computes segmentation of glioblastoma by its pre-trained AGU-Net models. For preoperative scans it provides a model that produces essentially the same ("BRATS-compliant") ET, ED, and NCR compartments as in the UCSF-PDGM dataset. For postoperative scans, it provides, in the case of GBM, a residual tumor segmentation model that gives a single-valued (one-compartment) mask as output. (ii) Our pipeline performs direct segmentation of the native T1 volume without registration to an atlas template, using the `recon-all-clinical` pipeline implemented in the recent version of the open-source `Freesurfer` software suite. The pipeline incorporates `SynthSeg` for segmentation of brain MRI scans of any contrast and resolution without retraining, and `SynthSR`, that turn heterogeneous clinical brain scans into high-resolution T1-weighted images for 3D morphometry. In effect, these tools provide similar segmentation results as the *virtual brain grafting* technique by Radwan and coworkers [81] (https://github.com/KUL-Radneuron/KUL_VBG) enabling whole-brain parcellation in the presence of large lesions using classical segmentation methods such as the `recon-all` pipeline in `Freesurfer`. For obtaining tumor location profiles of cortical and subcortical structures, `Raidionics` uses (diffeomorphic) imaging registration to a collection of standardized atlases in MNI space, covering cortical and subcortical regions with different granularity. This approach typically results in labeled regions that are only roughly aligned to the patient-specific sulci and gyri of the cortical sheet, including gray matter, white matter, and CSF, and only roughly approximating the fine details in subcortical brain structures of the underlying and the neighboring anatomy of tumor growth. (iii) The canonical segmentation engine for all reported subjects is the MONAI Bundle BraTS three-compartment SegResNet (Methods § *Evaluation framework*), with `segment_glioma` and `Raidionics` as compatible alternatives. The pipeline enables substitution with any state-of-the-art or new brain tumor segmentation method [4, 15, 57, 64, 70, 82, 90, 92, 106]. The engine runs behind a thin Python adapter that decouples the choice of backbone from downstream Hit-Plot code. Per-subject Dice, Hausdorff-95, and volumetric error of the unified engine against the dataset-provided reference masks are reported in Table 3. Hit-Plot-GT versus Hit-Plot-DL agreement is reported per cell as a quantitative measure of the Hit-Plot's sen-

sitivity to segmentation variability. A head-to-head benchmark of the unified engine against `Raidionics` and `segment_glioma` on the same $n = 50$ cohort is methodologically non-trivial: `Raidionics` is an atlas-warped pipeline operating in MNI space whereas HPGS evaluates in native subject space, so a single-cohort comparison would either resample `Raidionics` masks back to native space (and bias `Raidionics` downward via interpolation artefacts) or move HPGS to MNI space (abandoning the native-space architecture argued for elsewhere in the paper); and the preoperative `segment_glioma` model is trained and deployed on the PICTURE preprocessing contract (multiparametric MRI rigidly registered to T1c, N4-corrected, skull-stripped, and affine-registered to SRI-24), whereas HPGS evaluates on the dataset-provided native co-registered UCSF-PDGM volumes without re-running that pipeline, so a same-cohort paired comparison would confound segmenter identity with preprocessing and operating-space differences rather than isolate backbone quality. We leave a methodology-matched comparison to future work. As an additional bounded sensitivity analysis, we therefore ran `mri_TumorSynth` / nnU-Net v1.7 on the same $n = 50$ staged mpMRI inputs and report the paired voxel-space metrics (Table 6, Figure 17) and functional-anatomy Hit-Plot summary (Figure 18) in Appendix 5. The `TumorSynth` documentation expects skull-stripped inputs registered to the SRI-24 template, whereas this input-matched comparison deliberately uses the same staged mpMRI inputs as the MONAI Bundle run. Under these matched-input conditions, Appendix 5 measures the pipeline’s sensitivity to the segmentation backbone rather than `TumorSynth`’s best-case accuracy.

A pre-registered outlier audit on the Table 3 metrics flagged 16/50 subjects against the BraTS-2018 strict operating-point band ($WT \geq 0.85$, $TC \geq 0.75$, $ET \geq 0.70$ Dice; $HD95 > 20$ mm on any compartment). Visual adjudication using canonical-RAS T1c/reference/DL panels retained all sixteen: the dominant flag mode is the MONAI Bundle’s known mild under-segmentation of the enhancing-tumor boundary (DL/reference ET-volume ratios in the 0.55–0.90 range for 15/16 flagged subjects, with WT and TC ratios mostly in 0.85–1.10), and no flagged case meets a pre-registered hard-failure rule (lobar miss, contralateral hallucination, or artefact-dominated mask). Table 3 is therefore reported over the full $n = 50$ cohort with no QC-driven exclusions. The single substantive boundary-disagreement case is documented separately under Limitations (§ *Limitations and translational scope*, item vi). Using a multicompartment tumor segmentation, as in our case, facilitates quantitative assessment of the relationship between tumor enhancement, edema, partial T2-FLAIR mismatch sign, IDH mutational status, MGMT promoter methylation, and survival in glioblastoma, which might be of interest in large-scale studies [24, 60, 79].

Strengths and challenges

Using the `SynthSeg` algorithm, producing detailed anatomical parcellations, covering the whole intracranial 3D space, together with high-quality tumor segmentation that provides the boundaries of the complex, subject-specific geometry of tumor compartments in 3D, their intersection (*anatomical parcellation* $\cap$ *tumor region*) will have great potential for spatio-temporal characterization of brain tumor and tumor growth. For each of these segmented regions and their intersections, the corresponding spatially restricted multiparametric signal intensity distributions can be integrated with other methods and software tools, such as *radiomics* ([44]), *tumor growth modeling* ([68]), or radiation therapy *planning*. One challenge for our anatomical profiling features using mpMRI consisting of (T1, T1c, T2, FLAIR) only, is that they do not capture information processing neuronal substrate specifically, or the functional and structural connectivity between different parts of the brain - the regions and networks that are active during specific tasks (e.g., perception, motor, speech, working mem-

ory, reasoning). Before resecting a glioblastoma, or in photon or proton therapy, it can be crucial to know structural connections and functional areas around the tumor, especially if it is located near critical regions like those responsible for language or motor function. Diffusion MRI with fiber-tracking ([104]) and BOLD-fMRI with task-related or resting-state activations ([75]) can help map these regions, potentially aiding in preserving them during surgery or radiotherapy.

Future perspectives

Building on the recent deep learning approaches to neuro-oncological imaging, the present methods and case studies introducing anatomical localization and profiling as shown in this feasibility study, might also influence the evolution of RANO criteria [59, 73, 101, 107] for accurate and reproducible radiologic assessment in patients with glioblastoma. A possible follow-up of our approach is to more extensively address and evaluate our pipeline to the large-scale longitudinal LUMIERE dataset, which is well documented ([88]) and openly available data (on figshare.com) and code (<https://github.com/ysuter/gbm-data-longitudinal>). On a larger multi-centre grade-4 glioma cohort, our group has recently shown that deep-learning-derived tumour-compartment volumetrics and anatomical tumour trajectories carry independent prognostic information and may inform personalized radiotherapy margin determination [42], illustrating the outcome-coupled analyses that the present feasibility pipeline is intended to enable.

There is also accelerated research within *imaging-based computational oncology* regarding integrating data-driven methods (machine learning) with model-driven approaches (biophysical priors and statistical inversion) for predictive modeling of brain tumor growth, cf. Mang et al. [68] and the Dagstuhl Seminar “Inverse Biophysical Modeling and Machine Learning in Personalized Oncology” (<https://www.dagstuhl.de/23022>). In this context, Ezhov and coworkers [33] have recently introduced a deep learning-based methodology for inferring patient-specific spatial distribution of brain tumors from T1-Gd and FLAIR MRI scans with calibration of their partial differential equation (PDE) models of glioblastoma growth causing deformation of brain tissue (“mass effect”). The growth (forward) models were based on PDEs that describe the evolution of tumor cell density in the brain anatomy. The inverse model aims to identify free parameters of the forward model that best matches the observation, e.g., tumor outlines from medical imaging and tumor-surrounding tissue types, anatomical regions, and barriers. In this framework, detailed anatomical mapping of tumor heterogeneity and tumor environment for use as spatial conditions and parameter estimates can provide better mathematical modeling of glioma invasion. Such computational neuro-oncology approaches to personalized medicine involving spatiotemporal assessment and prediction of brain tumor growth has opened new avenues of research, [5, 33, 45, 62, 63, 86], where detailed anatomical tumor profiling will be important.

As the field of artificial intelligence increasingly turns its attention to generative AI and foundation models, neuro-oncological AI is following suit. In the future, we will likely see imaging data, their segmentations, anatomical profiling, metadata (tabular), and the referral information (text) as a single *multimodal document*. Based on the emergence of multimodal foundation models [74], this perspective is also expressed within the field of bioimage analysis [67, 84]. Incorporating image data, free text descriptions, and quantitative imaging-derived tumor biomarkers into large language models (LLMs) will enable powerful, case-based Q&A applications that can be instrumental for multidisciplinary diagnostics and treatment-planning teams [23] and also facilitate patient involvement and understanding.

A final observation and comment to our anatomical profiling endeavor is related to the highly influential

Brain Tumor Segmentation (BraTS) challenges, which have focused on the generation of a benchmarking environment and dataset for the delineation of adult brain gliomas since 2012 [70, 71]:

... For brain tumor patients, the image acquisition time series typically starts with an already pathological scan. This poses problems, as many algorithms are designed to analyze healthy brains and provide no guarantees for images featuring lesions. Examples include but are not limited to algorithms for brain anatomy parcellation, tissue segmentation, and brain extraction. To solve this dilemma, we introduce the BraTS inpainting challenge ...

These statements further indicate the importance and relevance of our contribution in the direction of computational oncology and precision imaging.

Limitations and translational scope

The present study is a feasibility and infrastructure report: it establishes a reproducible, native-space anatomy-profiling pipeline and quantifies its segmentation and Hit-Plot behaviour, but it does not test clinical utility. We therefore explicitly delimit the claims it can support. (i) **Cohort and generalisability.** The evaluation cohort is n=50 GBM (IDH-wildtype, WHO grade 4) subjects from a single public dataset (UCSF-PDGM v5, Sex $\times$ OS-tertile stratified, leak-safe with respect to BraTS21 Training, see Methods § *The UCSF-PDGM dataset* and Table 1). Although the cohort is sufficient to characterise the pipeline’s per-compartment segmentation and parcellation behaviour, multi-centre, prospective, treatment-stratified data (ideally including paired baseline / post-operative / follow-up exams from independent institutions) will be required before any generalisability statement to routine neuro-oncology practice can be made.

(ii) **Radiotherapy claims.** The phrase “sculpt dose distributions” (Introduction, in reference to modern IMRT/VMAT delivery [76, 77]) describes the capability of the radiotherapy delivery technology, not of our pipeline. Our pipeline contributes a more detailed, subject-specific anatomical substrate over which a treatment planner could in principle reason. It does not itself generate, modify, or validate dose distributions, and we have not performed any dosimetric or treatment-planning evaluation in this work. A prospective evaluation against a dosimetric end-point (e.g. NTCP for a defined organ-at-risk such as the contralateral hippocampus) is the appropriate next step and is outside the scope of the present manuscript.

(iii) **Margin-reduction claims.** The ESTRO–EANO 2023 reduction of the post-operative CTV margin from 20 mm to 15 mm cited in the Introduction is a clinical guideline change reported by Niyazi et al. [76]. We do not claim that our pipeline justifies, enables, or quantifies a further reduction of the CTV/PTV margin in any patient. Investigating whether anatomy-aware target-volume editing (e.g. restricting CTV expansion at known anatomical barriers identified by `recon-all-clinical` parcellations) can be performed safely will require a dedicated planning study with explicit recurrence-pattern analysis.

(iv) **Translational and outcome claims.** We do not report any patient-outcome end-point (overall survival, progression-free survival, neurocognitive performance, quality-of-life). The Hit-Plot / regional-tumour-burden representation is descriptive and is presented as an exploratory anatomical profile suitable for hypothesis generation, not as a clinically actionable biomarker. This applies equally to the functional-anatomy appendix summary and its OS-tertile small multiples: they show readability and metadata stratification of the same Hit-Plot quantities, not an association between regional burden and survival. Statements in the Introduction and Discussion concerning “preserving cognitive function” or “new therapeutic strategies” refer to the broader

literature on tumour–neuron interactions (e.g. [55, 46]) and are not conclusions of this study. A formal translational claim would require: (a) prospective, multi-centre validation of the segmentation and parcellation against expert manual reference for each compartment with standardised metrics (Dice, HD95, volumetric error, sensitivity, specificity; reported in Table 3 for our evaluation cohort); (b) a planning- or outcome-coupled study with predefined end-points; and (c) a regulatory pathway appropriate for the intended clinical use.

(iv-bis) **Hit-Plot-to-outcome as a named follow-up.** The UCSF-PDGM metadata summarised in Table 1 include overall survival, OS tertile, MGMT promoter status, and extent of resection for all $n = 50$ subjects of the evaluation cohort. A natural follow-up question is whether the per-subject Hit-Plot, or a low-dimensional summary of it (hemispheric asymmetry, lobar burden, or functional-anatomy group burden, optionally stratified by compartment), carries prognostic information beyond known clinical covariates (age, EOR, MGMT, treatment intensity). We deliberately do not perform that analysis in the present feasibility report. Under the same metadata filters used here (GBM, IDH-wildtype, WHO grade 4; known OS; no biopsy before imaging; exclusion of BraTS21 Training subjects), the UCSF-PDGM metadata support an expanded leak-safe pool of approximately 110 baseline subjects, with roughly balanced OS tertiles and a larger event count. That pool is a sensible next dataset for a pre-specified, low-dimensional regional-burden analysis, but using it would require re-running the segmentation, parcellation, Hit-Plot, QC, and manuscript-generation chain and is therefore a possible dedicated follow-up study. An honest test would require: a pre-registered design with dimensionality reduction applied before any per-region claim; a regularised Cox proportional-hazards model with age, EOR, MGMT, and treatment-intensity covariates forced in; FDR control over the remaining anatomical predictors; internal cross-validation; and external validation on an independent multi-centre cohort (e.g. BraTS21 Survival, UPenn-GBM, or a prospective multi-institutional set). At $n = 50$ subjects with ~ 300 candidate (region, compartment) features and ~ 30 observed events, the present cohort is statistically underpowered for a primary survival analysis, and even the larger UCSF-PDGM-only derivation pool would not by itself support a generalisability claim. The *Hit-Plot-to-outcome* variant is the natural follow-up this analysis points to.

(v) **Methodological scope.** The pipeline is built from established open-source components (the MONAI Bundle BraTS three-compartment SegResNet for tumor segmentation, the unified engine of Methods § *Evaluation framework*; FreeSurfer 8.2.0 `recon-all-clinical` for anatomical parcellation, with SynthSeg and SynthSR as its contrast-/resolution-agnostic core models; HD-BET for skull stripping; HD-GLIO-AUTO and DeepBraTumIA retained as comparators against the unified engine on the LUMIERE longitudinal subject, Table 5). The contribution of this work lies in the integration, the anatomy-aware Hit-Plot representation, and the reproducible end-to-end orchestration. The same unified engine plus FreeSurfer 8.2.0 `recon-all-clinical wmparc` pipeline drives both the cohort-level DL-vs-GT Hit-Plot agreement panel for the $n = 50$ UCSF-PDGM cohort (Figure 2) and the longitudinal Patient-048 Hit-Plot (Figure 8), so the same parcellation- $\cap$ -segmentation framework is exercised end-to-end on both the leak-safe cross-sectional cohort and the longitudinal arc without methodological substitution. Discriminating the relative contribution of each component to downstream features will require ablation studies. A methodology-matched head-to-head benchmark of the unified engine against `Raidionics` and `segment_glioma` (i.e. on a cohort, modality set, and spatial frame of reference where each comparator is on its own methodological home ground) is another standing future-work item.

(vi) **Reference-mask quality and cross-segmenter disagreement.** The UCSF-PDGM tumor masks used as reference for the $n = 50$ segmentation evaluation (Table 3) are themselves the output of a specific human-

plus-algorithmic annotation protocol and are treated as a *reference* rather than ground truth. The pre-registered outlier audit reported in Methods (§ *Per-subject outlier audit on Table 3*) identifies one subject (sub-0396) as a worst-case exemplar of cross-segmenter boundary disagreement: the unified MONAI Bundle pipeline labels a confluent necrotic-core + enhancing-rim region (~ 17 cc) within the central lesion, whereas the dataset-provided reference labels the same region predominantly as edema (reference TC ~ 2 cc on a ~ 88 cc total lesion). Without expert clinical adjudication beyond the scope of this study, we cannot determine which segmentation is closer to the underlying tumour biology. The case is rendered as a standalone T1c/reference/DL panel in the supplementary material (produced by the same canonical-RAS pipeline used for Figure 2), and sub-0396 is retained in Table 3 to keep the reported distribution faithful to the true cross-segmenter behaviour on this cohort.

(vii) **Per-region sunbursts and functional-anatomy label groups.** Per-region anatomical-profile sunbursts at single-subject and cohort-stratified levels are descriptive at $n=1$ and aggregate poorly when collapsed to coarse anatomical lobes (which discards the per-region resolution that the rest of the Hit-Plot variants are designed to expose). The cohort-level lobe distribution is therefore reported as a quantitative prose statement in Methods § *The UCSF-PDGM dataset* rather than as a sunburst. The compact functional-anatomy summary in Appendix 4 is a more readable aggregation, and is included because it fits as one descriptive figure plus one cautious Results paragraph. Appendix 5 repeats the same aggregation after swapping the segmentation backbone to `mri_TumorSynth / nnU-Net v1.7` as a bounded sensitivity check. Both appendices remain anatomical-label adjacency analyses derived from `wmparc`: the motor-adjacent, left-language-adjacent, and deep-midline / commissural groups are not tractography, individualized eloquence mapping, fMRI, organ-at-risk proximity, treatment-planning validation, or outcome prediction. Proper per-region sunbursts, tract / organ-at-risk proximity profiles, and Hit-Plot-to-outcome models are deferred to follow-up studies where cohort size, clinical annotations, and independent validation can support those stronger claims.

5 Code and data availability

The UCSF-PDGM dataset used in this study is publicly available from The Cancer Imaging Archive (TCIA) (<https://doi.org/10.7937/tcia.bdgm-8v37>). The LUMIERE dataset is publicly available at <https://github.com/ysuter/gbm-data-longitudinal>. All processing code, the pinned software stack, the cohort-pinning manifests for the $n = 50$ UCSF-PDGM evaluation cohort and for the six-timepoint LUMIERE Patient-048 longitudinal subject, and the producer scripts that generate every figure and table reported above end-to-end from the on-disk derivatives will be released at <https://github.com/arvidl/high-precision-glioma-segmentation>. The qualitative LUMIERE Patient-048 panel of Figure 7 was produced manually in Freeview on FreeSurfer 7.4.1 `recon-all-clinical` outputs. The panel-composition recipe is available from the corresponding author on request.

Author contributions

Conceptualization: A.L. and S.A.; methodology: A.L., A.S.L., and S.A.; software: A.L., A.S.L., and S.A.; validation: A.L. and S.A.; formal analysis: A.L. and A.S.L.; investigation: A.L., A.S.L., S.A., and M.H.H.; resources: A.L., A.S.L., and M.C.; data curation: A.L.; writing, original draft: A.L.; writing, review and

editing: A.L., A.S.L., S.A., D.G., M.H.H., and M.C.; visualization: A.L. and S.A.; supervision: A.L., A.S.L., and M.C.; project administration: A.L.; funding acquisition: A.L., A.S.L., and M.C. All authors reviewed the manuscript and agreed to its submission.

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Institutional review board statement

The study was conducted in accordance with the Declaration of Helsinki. For the UCSF-PDGM dataset, imaging studies were approved by the University of California, San Francisco institutional review board with a waiver of informed consent, as described in the original dataset publication [22]. The LUMIERE dataset was collected under institutional review board approval at the University Hospital of Bern, Switzerland [88]. No additional ethical approval was required for the re-analysis of these publicly available, anonymised datasets.

Declaration of generative AI use

During preparation of this manuscript, the authors used Cursor (version 3.x) with the Composer 2.5 agent to assist with pipeline software development and language editing of the text. The tool was used to accelerate implementation and revision of analysis code and to improve clarity of the prose. It was not used to generate scientific results, figures, or reference lists without human verification. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of the published article.

Abbreviations

The following abbreviations are used in this manuscript:

ANTs	Advanced Normalization Tools (software toolkit)
ART	Adaptive radiotherapy planning
ASL	Arterial spin labeling (pMRI measurement technique)
BOLD	Blood oxygenation level-dependent (fMRI measurement technique)
BraTS	Brain Tumor Segmentation (challenges)
CCC	Concordance correlation coefficient
CSF	Cerebrospinal fluid
CTV	Clinical target volume
DCE-MRI	Dynamic susceptibility contrast MRI with the administration of a contrast agent
DR	Domain randomization
dMRI	Diffusion MRI
eTIV	Estimated total intracranial volume
fMRI	Functional MRI
ED	Tumor region with edema
EOR	Extent of resection
ESTRO	European Society for Radiotherapy & Oncology
ET	Enhancing tumor region
FA	Fractional anisotropy
FLAIR	Fluid attenuated inversion recovery (MRI measurement technique)
GBM	Glioblastoma
GSI-RADS	Glioblastoma Surgery Imaging Reporting And Data System
GTR	Gross total resection
GTV	Gross tumor volume
HARDI	High-angular-resolution diffusion imaging (MRI measurement technique)
HD ₉₅	95th-percentile Hausdorff distance
HGG	High-grade glioma
ICV	Intracranial volume
IDH	Isocitrate dehydrogenase (enzyme)
IMRT	Intensity-modulated radiotherapy
MGMT	O6-methylguanine-DNA methyltransferase (enzyme)
MNI	Montreal Neurological Institute
mpMRI	Multiparametric MRI
NCR	Necrotic tumor region
NET	Non-enhancing tumor region
NTCP	Normal tissue complication probability (models)
OAR	Organs at risk
OS	Overall survival
PCR	Polymerase Chain Reaction
PTV	Planning tumor volume
RANO	Response Assessment in Neuro-oncology (criteria)
scRNA-seq	Single-cell RNA sequencing
STR	Subtotal resection
SWI	Susceptibility-weighted imaging (MRI measurement technique)
SyN	Symmetric normalization: Affine + deformable transformation
pMRI	Perfusion MRI
RT	Radiotherapy
TC	Tumor core
TCGA	The Cancer Genome Atlas
TCIA	The Cancer Imaging Archive
TME	Tumor microenvironment
UCSF-PDGM	University of California San Francisco Preoperative Diffuse Glioma (dataset)
WT	Whole tumor or Wild type

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The authors report there are no competing interests to declare.

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APPENDIX 1. Examples of image data from the UCSF-PDGM and LUMIERE datasets

UCSF-PDGM

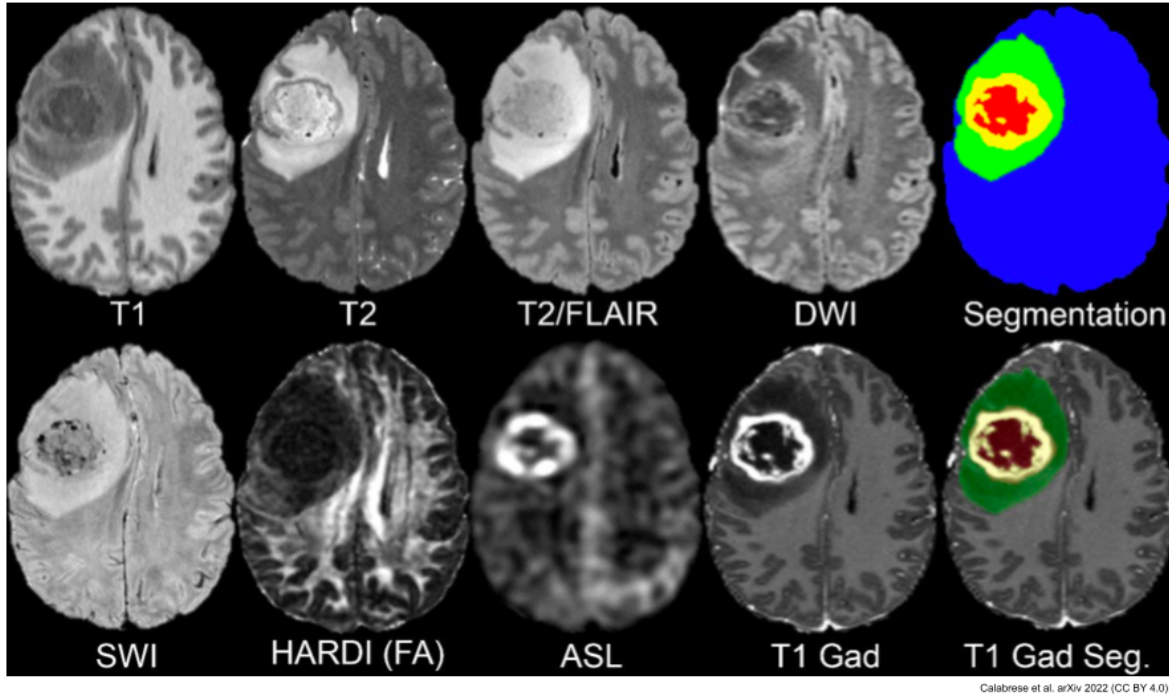


Figure 9: **Example of multiparametric MRI examination in a 37-year-old man from the UCSF-PDGM dataset.** The original recordings have been subject to non-linear co-registration, resampling, and skull stripping. T1: T1-weighted pre-contrast, T2: T2-weighted, T2/FLAIR: T2-weighted FLAIR, DWI: isotropic (trace) diffusion-weighted image, SWI: susceptibility-weighted image, HARDI (FA): fractional anisotropy derived from high-angular-resolution diffusion imaging (HARDI) data, ASL: arterial spin labeling perfusion, T1 Gad: T1-weighted post-contrast, Segmentation: multicompartment tumor segmentation (blue = brain, green = FLAIR abnormality, yellow = enhancing tumor, red = necrotic core), T1 Gad Seg.: tumor segmentation semitransparent overlay on T1-weighted post-contrast image. We used the mpMRI subset {T1, T1 Gad, T2, T2/FLAIR, Segmentation} for the present study.

From Calabrese et al. [22] <https://arxiv.org/abs/2109.00356> (CC BY 4.0).

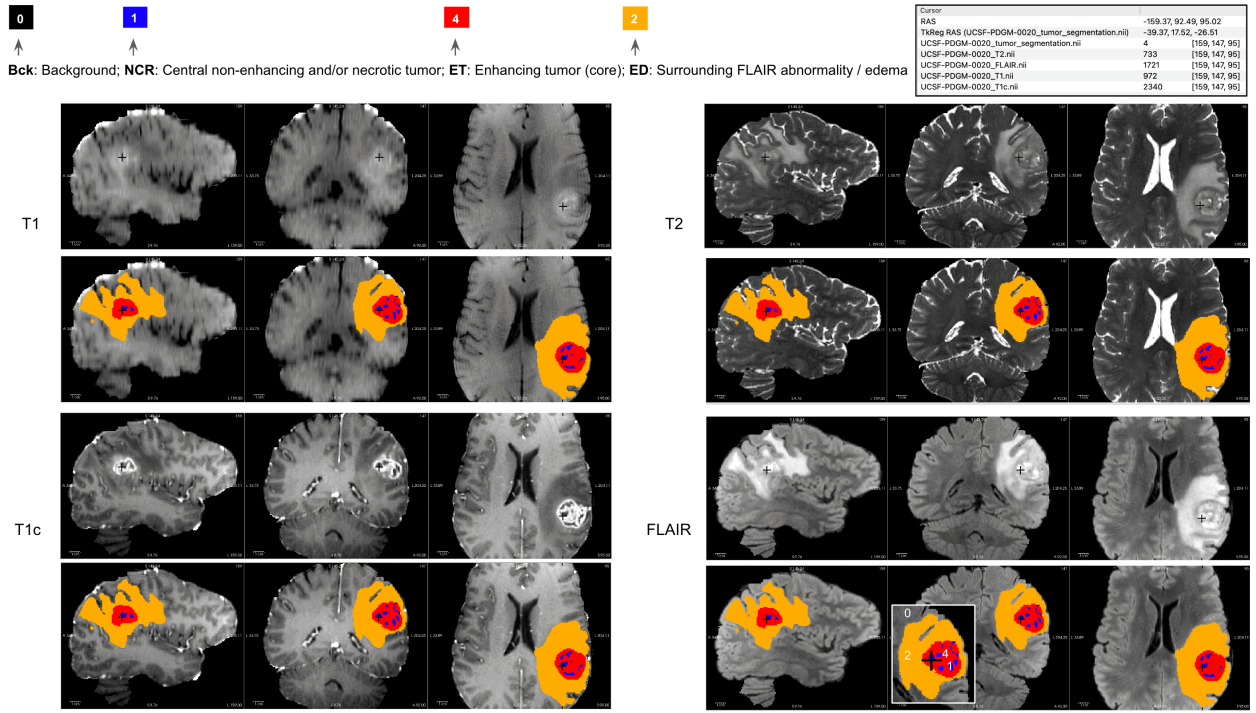


Figure 10: **Illustration of the mpMRI data and tumor segmentation using the UCSF-PDGM-0020 examination.** Each of the four channels (T1, T1c, T2, and FLAIR) is shown separately as rows in the panels, with the corresponding color-coded three-compartment tumor segmentation mask as an overlay, shown below. A sagittal, a coronal, and an axial slice are the columns in the panels. The center of the black crosshair is fixed at coordinates $[i=159, j=147, k=95]$ in all panels, using the interactive `Freeview` tool. The mapping of tumor component labels (NCR, ED, ET) to integer values (1, 2, 4) and color-codes is shown in the upper panel.

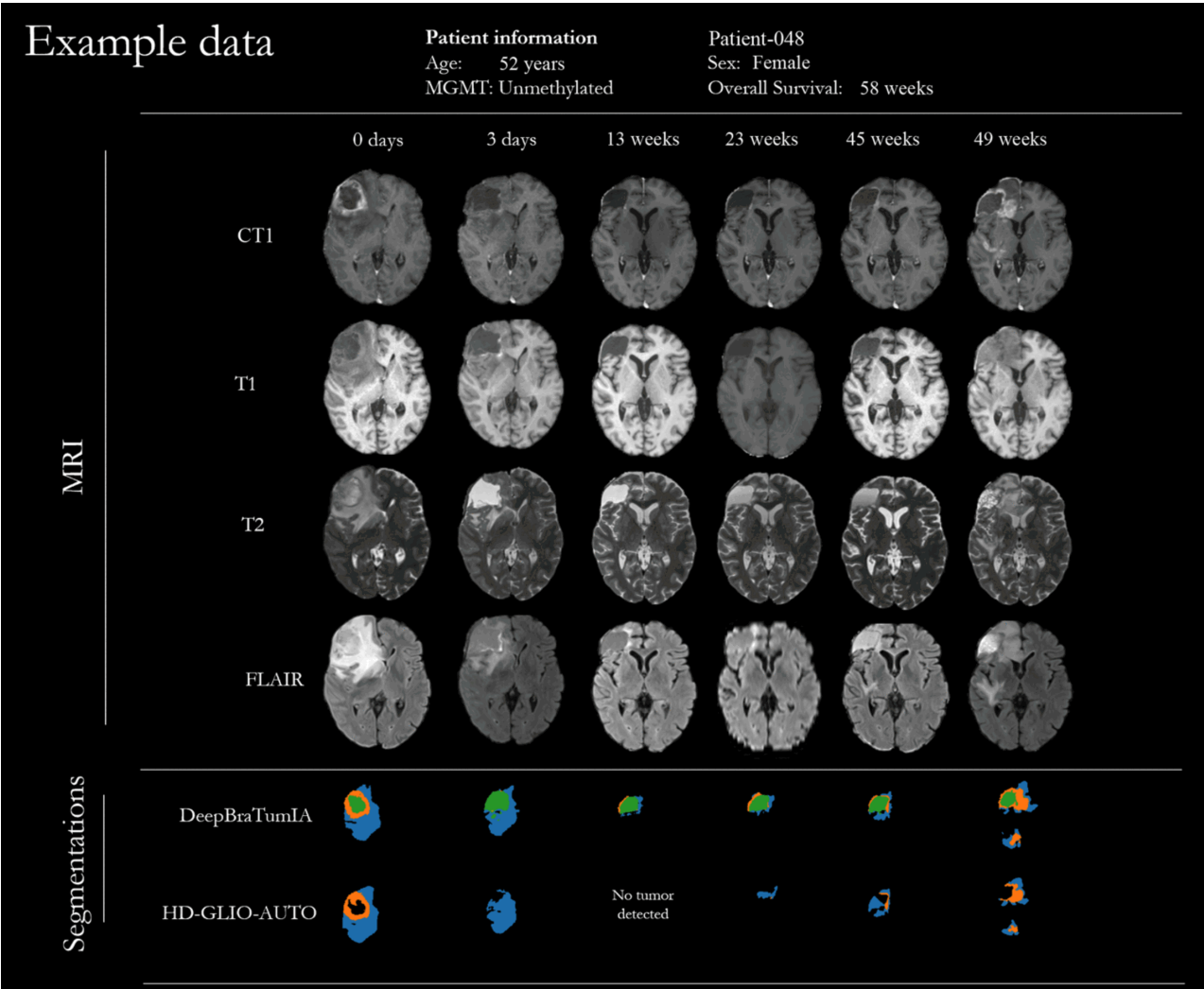


Figure 11: **Example of longitudinal data on GBM from the LUMIERE collection.** CT1/T1: Post-/pre-contrast T1-weighted, T2-weighted, and fluid-attenuated inversion recovery (FLAIR). For this individual, the entire contrast enhancement was fully removed. Following a phase of complete response, there was a progression characterized by a highly rapid growth rate. In the present study we process all six mpMRI examinations of Patient-048 under the unified evaluation framework (Results, Figure 6).

From Suter et al. [88] <https://www.nature.com/articles/s41597-022-01881-7> (CC BY 4.0).

APPENDIX 2. Anatomical profiling results from cases UCSF-PDGM-0039, -0066, and -0085

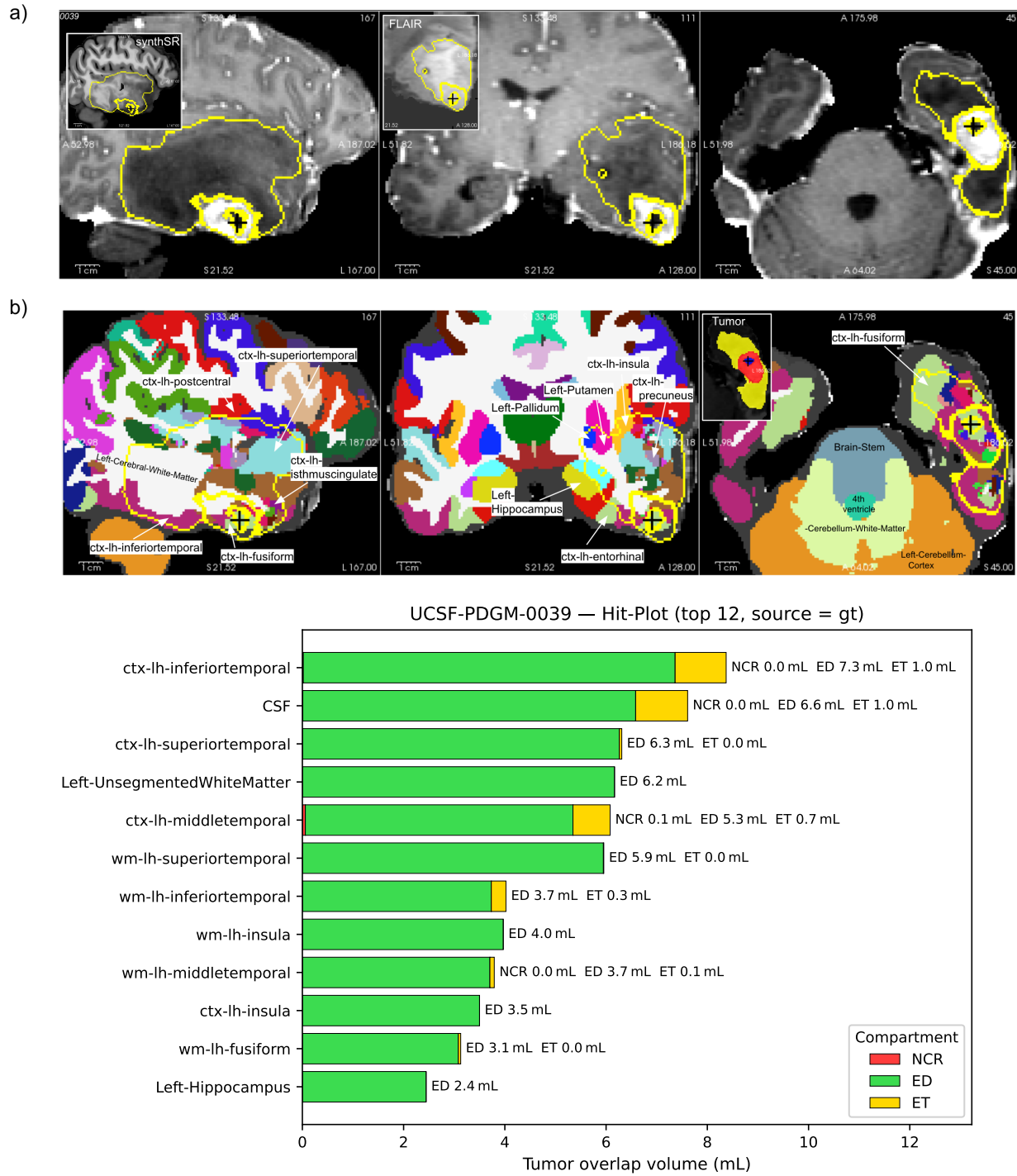


Figure 12: Tumor location and anatomical neighborhoods from the study UCSF-PDGM-0039. a) The original Gadolinium-contrast enhanced T1-weighted image (T1c) with an overlay of tumor region boundaries. Insert in the left panel shows the corresponding synthesized super-resolution T1 MPRAGE image (synthSR) in a para-sagittal section. Insert in the middle panel, covering the tumor in a coronal section, showing the corresponding FLAIR channel. b) The `synthseg` (not `wmparc`) segmentation with an overlay of tumor compartment boundaries and annotation of selected anatomical structures and regions in the tumor surroundings. Insert in the right panel shows the corresponding three-compartment tumor segmentation using our extended `FreeSurferTumorColorLUT` lookup table. The black crosshair center is located at $[i=167, j=111, k=45]$. Visualization of normal and abnormal tissue in the sagittal, coronal, and axial planes are made with `Freeview` and color-coded labels according to the `FreeSurferColorLUT` lookup table. c) Tumor compartment “Hit-Plot” for the top-12 `wmparc` parcels of this subject.

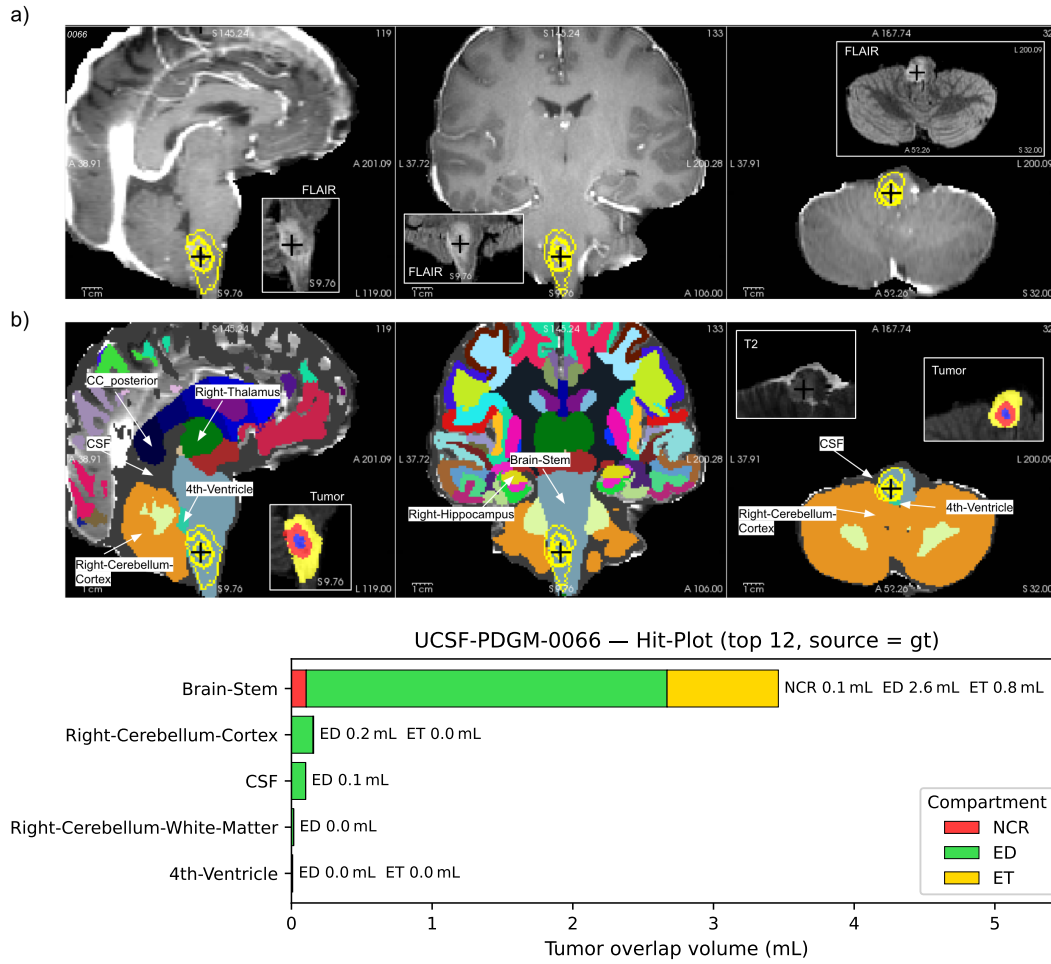


Figure 13: Tumor location and anatomical neighborhoods from the study UCSF-PDGM-0066. a) The original Gadolinium-contrast enhanced T1-weighted image (T1c) with an overlay of tumor region boundaries. Inserts covering the tumor region show the corresponding FLAIR channel without the superposition of tumor region boundaries. b) The *wmparc* segmentation with an overlay of tumor compartment boundaries and annotation of selected anatomical structures and regions in the tumor surroundings. Inserts in the left and right panel show the corresponding three-compartment tumor segmentation using our extended *FreeSurferTumorColorLUT* lookup table. The right panel also shows an insert of the T2 channel with the small tumor partly surrounded by CSF. The black crosshair center is located at $[i=119, j=133, k=32]$. Visualization of normal and abnormal tissue in the sagittal, coronal, and axial planes are made with *Freeview* and color-coded labels according to the *FreeSurferColorLUT* lookup table. c) Tumor compartment “Hit-Plot” for the top-12 *wmparc* parcels of this subject (this small lesion is partly surrounded by CSF, which appears in the bar chart as a non-trivial CSF compartment overlap).

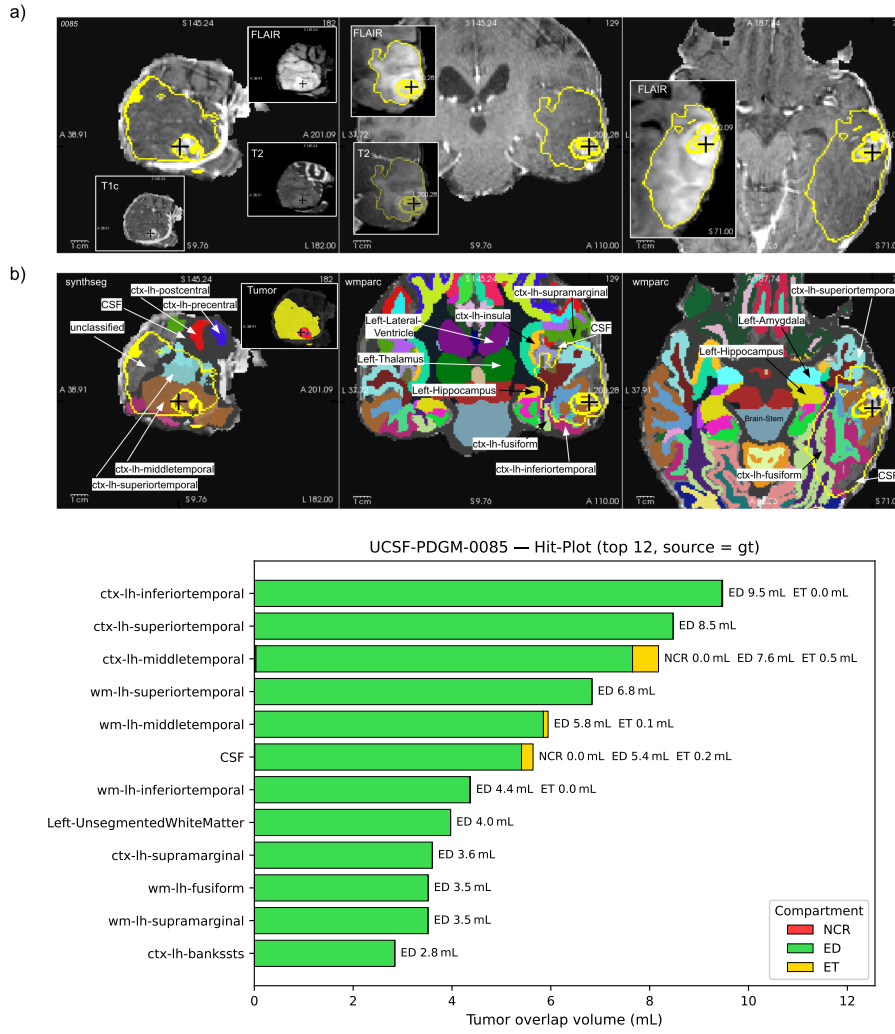


Figure 14: Tumor location and anatomical neighborhoods from the study UCSF-PDGM-0085. a) The original Gadolinium-contrast enhanced T1-weighted image (T1c) with an overlay of tumor region boundaries. Insert in the right panel shows the corresponding FLAIR channel. b) The *wmparc* segmentation with an overlay of tumor compartment boundaries and annotation of selected anatomical structures and regions in the tumor surroundings. Insert in the right panel shows the corresponding three-compartment tumor segmentation using our extended *FreeSurferTumorColorLUT* lookup table. The black crosshair center is located at $[i=182, j=129, k=71]$. Visualization of normal and abnormal tissue in the sagittal, coronal, and axial planes are made with *Freeview* and color-coded labels according to the *FreeSurferColorLUT* lookup table. c) Tumor compartment “Hit-Plot” for the top-12 *wmparc* parcels of this subject.

APPENDIX 3. Per-subject DL-vs-reference Hit-Plot agreement on five representative leak-safe cohort subjects

The five detail subjects of *APPENDIX 2* (UCSF-PDGM-0039, -0066, -0085) and Figures 4 and 5 are leak-unsafe for the unified BraTS21-derived DL engine by construction (four of five sit in the BraTS21 Segmentation *Training* cohort; the fifth in *Validation*). To answer the per-subject form of “how does the DL Hit-Plot compare to the reference Hit-Plot” without violating the leak-safe evaluation paradigm, the supplementary panel below uses five *representative* subjects drawn from the $n = 50$ evaluation cohort (cohort manifest cited in Methods) where DL inference is legitimate test-set inference.

Selection rule (S-D). The five subjects shown are those whose mean absolute per-cell residual, averaged across all (parcel, compartment) cells of the per-cell residual table computed for Figure 2, is closest to the cohort mean residual; ties are broken by subject id ascending. The pick is therefore deterministic and produces a representative slice of the cohort agreement distribution rather than a cherry-picked best/worst.

Per-subject DL-vs-GT Hit-Plot agreement (S-D rule, $n=5$ representative cohort subjects)

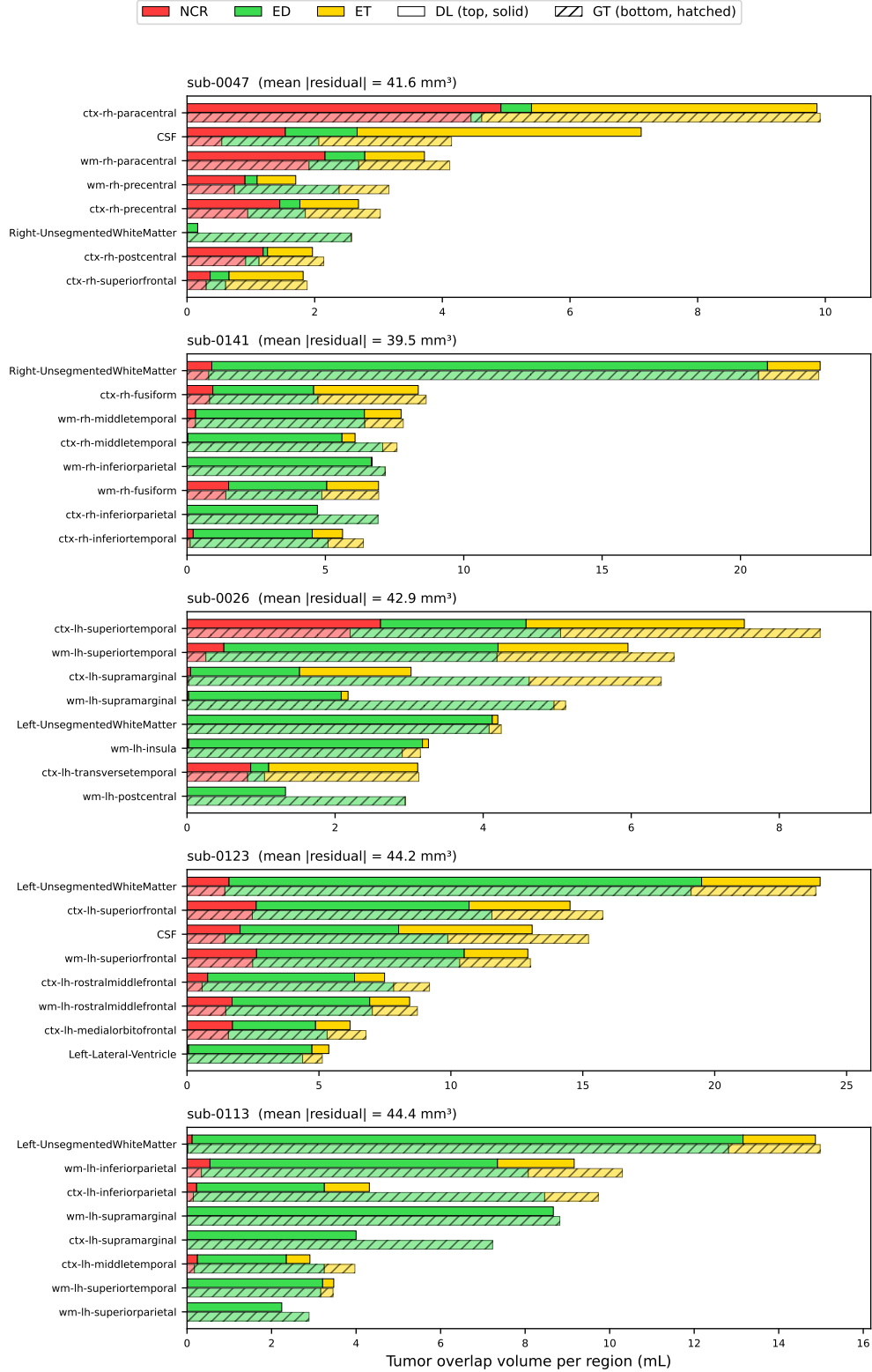


Figure 15: Per-subject DL-vs-reference Hit-Plot agreement on five representative subjects from the $n = 50$ evaluation cohort. Each row shows one subject; every region has two horizontal bars, the unified DL engine's Hit-Plot on top (solid colours, NCR + ED + ET stacked) and the dataset-provided reference Hit-Plot below (same colours, hatched). Top-8 $wmparc$ regions per subject are shown, ranked by the maximum of DL- and reference-side total overlap volume. Subjects are picked by the deterministic S-D rule (*Selection rule* above); the cohort mean absolute per-cell residual (41.07 mm³) and per-subject means appear in each subplot title. Compartment colours match Figure 3 panel (e).

APPENDIX 4. Functional-anatomy Hit-Plot summary on the leak-safe cohort

The panel below summarises the $n = 50$ UCSF-PDGM Hit-Plot-GT matrix after collapsing `wmparc` labels into a small set of pre-defined anatomical groups. The top-left panel uses the full cohort; the remaining panels repeat the same summary by OS tertile (short, mid, long) as an exploratory stratified view.

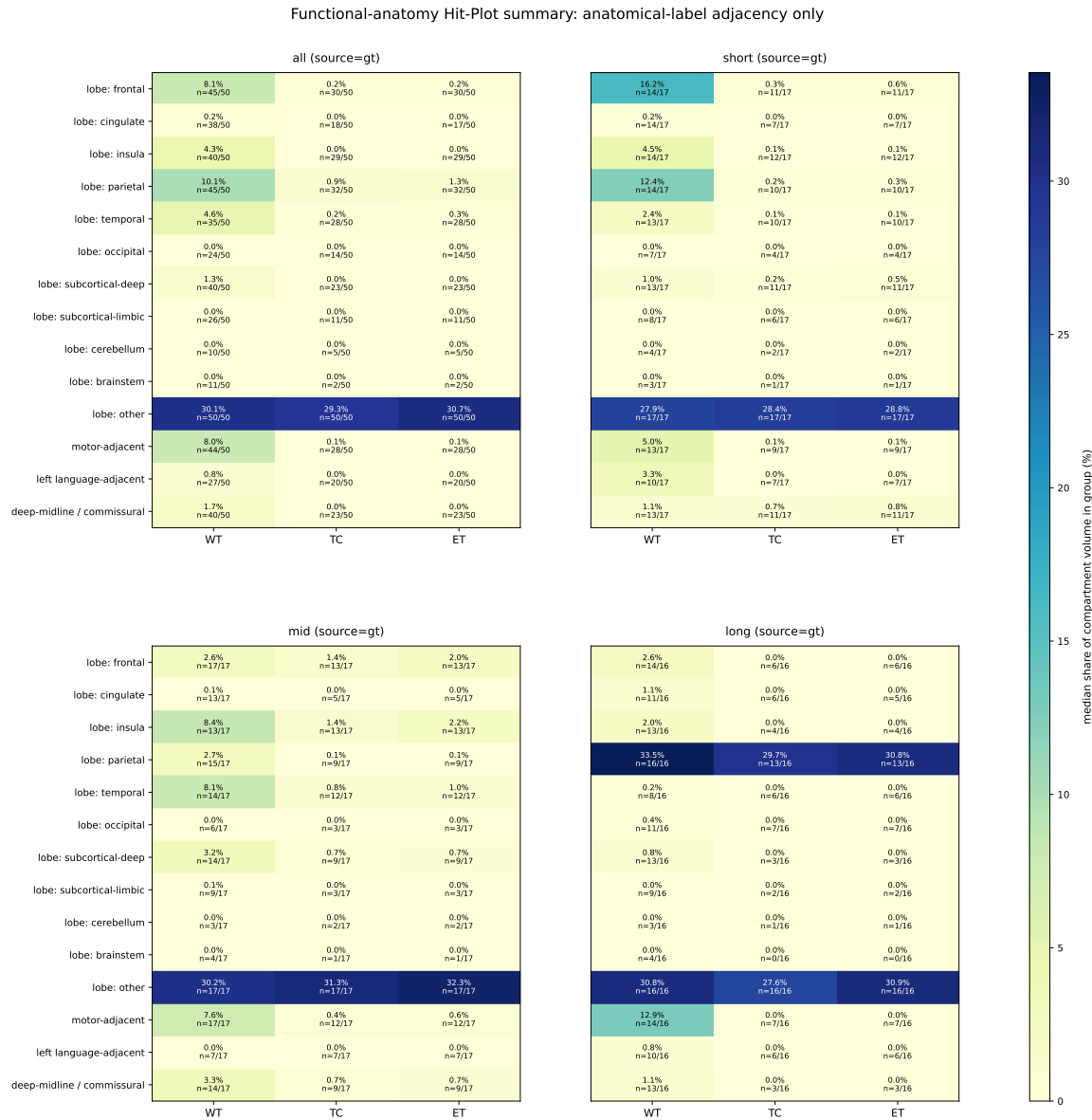


Figure 16: **Functional-anatomy summary of the Hit-Plot on the leak-safe UCSF-PDGM cohort.** Rows are pre-defined `wmparc`-derived groups: the existing lobar bins plus three anatomical-label adjacency summaries (motor-adjacent, left language-adjacent, and deep-midline / commissural). Columns are BraTS compartments (WT, TC, ET). Cell colour gives the median share of the compartment volume falling in that group; text gives the median percentage and the prevalence denominator $n = K/N$ for any overlap in the group. The figure is generated from the existing per-subject Hit-Plot CSVs and `wmparc` LUT sidecars; a companion DL sensitivity run gives similar all-cohort WT summaries (motor-adjacent 43/50, median 6.91 %; left language-adjacent 27/50, median 0.71 %; deep-midline / commissural 42/50, median 2.06 %). These groups are anatomical-label adjacency summaries only: they are not tractography, individualized eloquence mapping, fMRI, organ-at-risk proximity, treatment-planning validation, or outcome prediction.

Interpretation. Appendix 4 collapses high-dimensional Hit-Plots into readable anatomical-label groups while preserving their descriptive role: WT commonly involves motor-adjacent and deep-midline / commissural groups, whereas left language-adjacent overlap is lower-burden; OS-tertile panels are a readability check, not an outcome-association test.

APPENDIX 5. TumorSynth sensitivity analysis of the functional-anatomy Hit-Plot

The table and panels below repeat the Appendix 4 cohort-level Hit-Plot summaries after replacing the MONAI Bundle BraTS three-compartment SegResNet masks with `mri_TumorSynth` / nnU-Net v1.7 masks generated for all 50/50 leak-safe UCSF-PDGM subjects. TumorSynth here runs on the same staged mpMRI inputs as the MONAI Bundle, rather than the SRI-24-registered, skull-stripped volumes it expects; the comparison below is therefore a bounded check on backbone sensitivity, and should not be read as a best-case TumorSynth benchmark.

Table 6: **Voxel-space agreement between the unified MONAI Bundle segmentation and the `mri_TumorSynth` sensitivity run on the $n = 50$ UCSF-PDGM cohort.** Values are median [Q1, Q3] over subjects for the paired comparison DL vs. TumorSynth, using the same WT, TC, and ET compartment definitions as Table 3. The comparison is input-matched to the HPGS staged mpMRI files and is therefore interpreted as a bounded segmentation-backbone sensitivity analysis.

Metric	WT	TC	ET
Dice	0.894 [0.866, 0.921]	0.860 [0.793, 0.901]	0.672 [0.559, 0.736]
HD ₉₅ (mm)	4.06 [3.20, 5.83]	4.30 [3.04, 6.38]	4.00 [3.00, 5.39]
VE (mm ³)	6544 [3547, 12452]	2652 [906, 6561]	3704 [793, 7590]
Sensitivity	0.878 [0.831, 0.914]	0.894 [0.777, 0.946]	0.660 [0.507, 0.762]
Specificity	1.000 [0.999, 1.000]	1.000 [0.999, 1.000]	1.000 [0.999, 1.000]

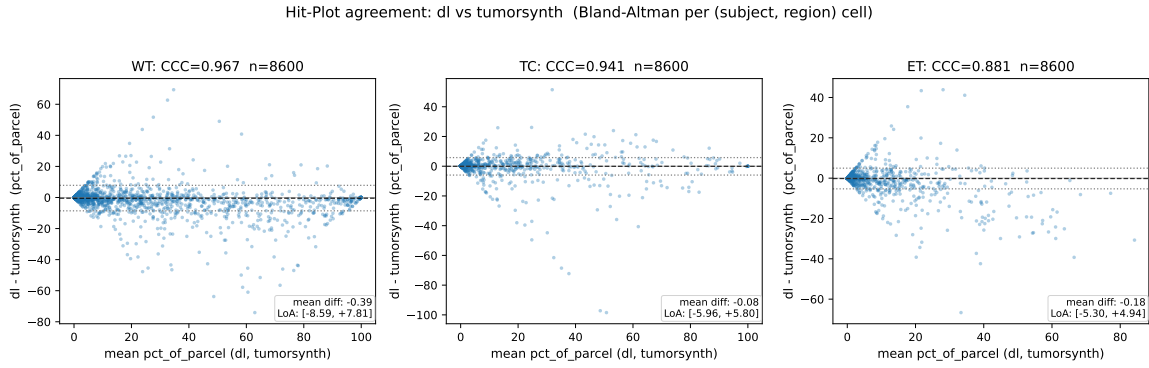


Figure 17: **Cell-level Hit-Plot agreement between the unified MONAI Bundle segmentation and `mri_TumorSynth`.** Each point is one subject-by-wmparc-region cell ($50 \times 172 = 8600$ cells per compartment). Bland-Altman summaries compare the DL and TumorSynth Hit-Plot percentage of parcel volume for WT, TC, and ET. Concordance remains high at the anatomical-profile level (CCC 0.967 WT, 0.941 TC, 0.881 ET), with mean differences of -0.39 , -0.08 , and -0.18 percentage points of parcel volume, respectively.

Functional-anatomy Hit-Plot summary: anatomical-label adjacency only

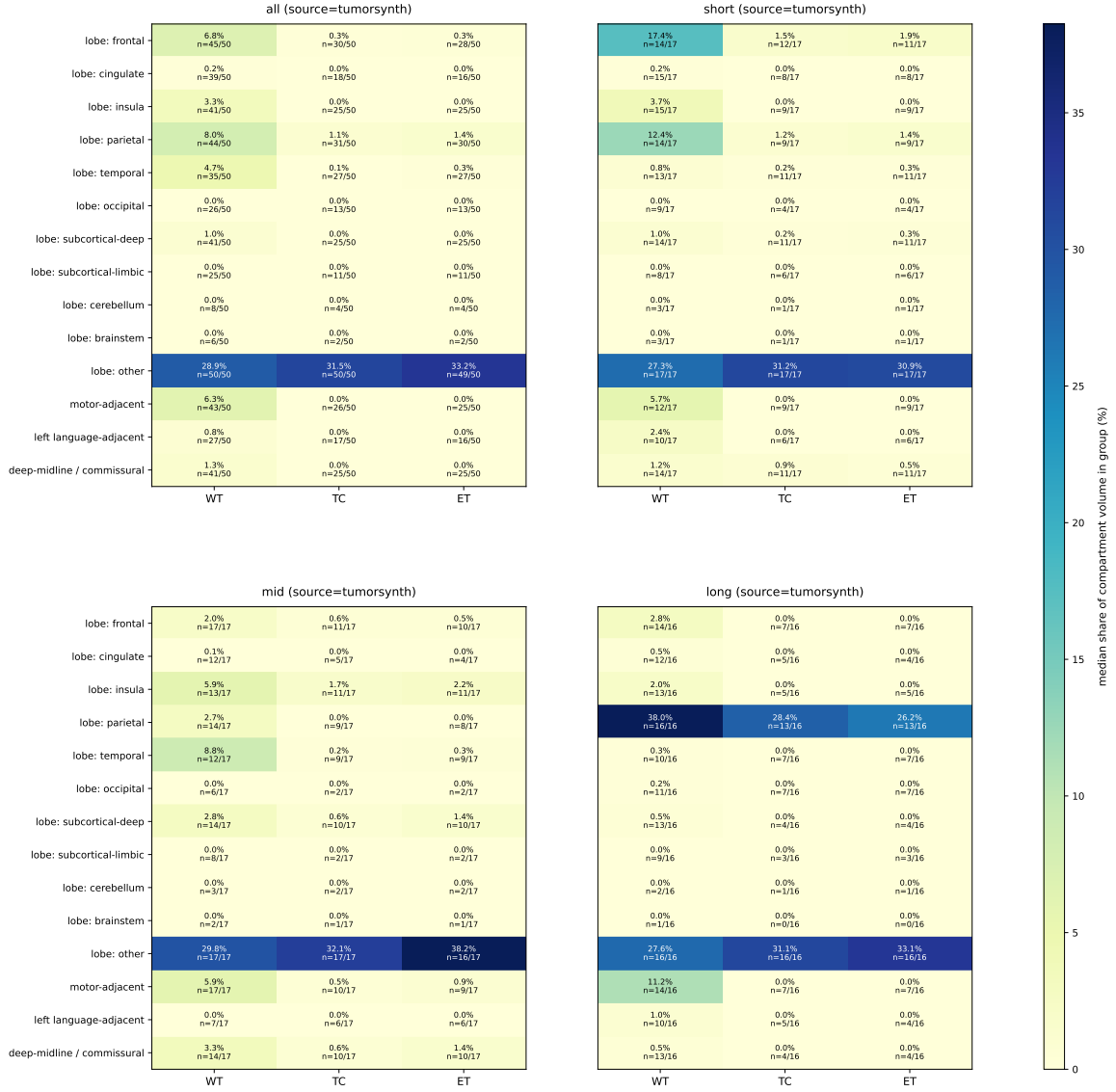


Figure 18: **Functional-anatomy summary of the `mri_TumorSynth`-driven Hit-Plot on the leak-safe UCSF-PDGM cohort.** The layout, groups, and cell semantics match Figure 16: rows are pre-defined wmparc-derived lobar and anatomical-label adjacency groups; columns are WT, TC, and ET; colour gives the median share of compartment volume in the group; text gives median percentage and prevalence $n = K/N$. The all-cohort WT summaries remain close to the reference-mask and MONAI Bundle summaries in Appendix 4 (motor-adjacent 43/50, median 6.27%; left language-adjacent 27/50, median 0.84%; deep-midline / commissural 41/50, median 1.28%), while voxel-space ET agreement is more variable (Table 6). This supports the use of the Hit-Plot as a readable anatomy-aware summary while preserving the methodological caveat that TumorSynth was not run under its ideal SRI-24-registered preprocessing regime.

Interpretation. Appendix 5 shows that the cohort-level anatomical profile is broadly preserved after swapping the segmentation backbone from the MONAI Bundle model to `mri_TumorSynth`: voxel-space agreement is strongest for WT and TC and weaker for ET, but the cell-level Hit-Plot concordance remains high and the functional-anatomy WT summaries retain the same qualitative pattern. We therefore interpret the appendix as evidence that the descriptive regional profile is reasonably stable to this bounded backbone perturbation, while maintaining the caveat that TumorSynth was not run under its ideal SRI-24 preprocessing regime.